

Granular Information and Sectoral Movements^{*}

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Abstract

This paper shows a strong link between the granular information contained in individual stock prices and sectoral movements. We find that a predictor aggregating the price movements of a broad cross section of individual stocks predicts sector ETF returns at intraday and lower frequencies. When we further incorporate the information from structural models, the resulting information signals have even stronger return predictability. A trading strategy that exploits the return predictability is profitable after trading costs. These results support theories of granular and network origins of aggregate shocks.

JEL classification: G10, G14, G40.

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1. Introduction

Firms operate in a complex network with intricate linkages. The COVID-19 pandemic pinpoints the importance of the interconnections among them. For instance, the disruptions of semiconductor chip producers sent ripples across numerous sectors in the economy, including the energy, healthcare, consumer electronics, and automotive industries, etc. Take the automotive industry as an example. It was benefiting from the surging demand for vehicles when the economy recovered from the COVID-induced collapse. The shortage of semiconductor chips, however, struck a heavy blow to the automotive producers, leading to a halt in manufacturing and to losses in operations.¹ This episode well illustrates the granular and network origins of sectoral and aggregate movements (Gabaix, 2011; Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi, 2012). In these theories, granular shocks generate aggregate fluctuations due to their own importance in the economy and through a propagation mechanism arising from economic links among firms.

Motivated by these theories, we explore the connections between the granular, firm-level information and sectoral fluctuations with stock market data. In particular, we use returns on individual stocks to capture firm-level information and returns on the sector exchange-traded funds (ETFs) to capture sectoral fluctuations. The study of sectoral fluctuations rather than univariate aggregate market fluctuations, and the inclusion of individual stocks throughout the economy give us more power to identify the rich connections.

To mitigate the concern of reverse causality, we focus on the predictive power of individual stock returns for the sector ETF returns. One empirical challenge for this analysis is that the

¹For instance, in March 2021, a fire at a Japanese semiconductor factory owned by Renesas Electronics Corporation further exacerbated the global chip shortage. The Japanese chip producer accounted for 30% of the global market for microcontroller units used in cars. This incident, among other concerns, led the IHS Markit to expect disruptions to the production of nearly 1.3 million global light vehicles (“Global auto recovery to take more hits from Japan chip plant fire, severe U.S. weather: IHS,” *Reuters*, March 31, 2021).

cross section of individual stocks has a very high dimension. As a result, standard regressions using individual stock returns to forecast sector returns can easily lead to overfitting and spurious statistical findings. To circumvent this problem, we use the LASSO shrinkage method along with other statistical techniques to reduce the dimensionality. Based on the predictive model for each sector ETF, we construct an ETF-level signal that aggregates firm-level information into sector-level information. Then we perform cross-sectional regressions using the resulting signal to predict future ETF returns.

Because the sample of sector ETFs covers a relatively short time series, in our initial analyses, we focus on high-frequency returns to increase the power of the tests. We start the analyses by training the LASSO signal using half-an-hour stock returns to forecast subsequent half-an-hour ETF returns. To predict the first half-an-hour ETF returns involving the market open, we use the overnight returns on individual stocks to generate the corresponding LASSO signal. The resulting signal has strong predictive power for future ETF returns. In Fama-MacBeth regressions, the slope coefficients for the LASSO signal range from 0.10 to 0.19 with an average of 0.14, the t -statistics have an average value of 8, and the adjusted R^2 's are above 14%. In these regressions, we control for the news momentum effects of the ETF constituent firms (Jiang, Li, and Wang, 2021). This result shows the power of using granular origins to identify sectoral shocks.

The plain-vanilla shrinkage method of aggregating firm-level information has superior ability in the ideal world when we have perfect and infinite amounts of data. In practice, however, this is never true. In a finite sample, the LASSO signal may be inefficient. Moreover, it does not take into account the insights from the structural economic linkages across firms. For instance, information tends to travel slowly across the supply chain (Cohen and Frazzini, 2008; Menzly and Ozbas, 2010); in a given sector, firms with high analyst coverage tend to be the bellwether, leading the sectoral movements (Hameed, Morck, Shen, and Yeung, 2015). Does incorporating structural information into the machine learning algorithm help to improve its performance?

To address this question, we develop a simple procedure that distills the structural information into the machine learning algorithm. First, for each sector ETF, we select firms that ex ante are connected to firms in the sector. For instance, to use the supply chain information, we rely on the Input-Output (IO) Table of the Bureau of Economic Analysis (BEA) to identify the customers and suppliers (C-S) of a given sector, i.e., the economically linked firms. Second, we apply the LASSO shrinkage method to statistically select firms within this pre-screened sample. We refer to the resulting ETF-level signal as the CS-based LASSO signal ($LASSO_{CS}$). Similarly, to capture the leading firm effect, for each sector ETF, we select high-analyst-coverage firms from the same sector, and then implement the LASSO shrinkage method. We refer to the resulting signal as the Leading-Firm-based LASSO signal ($LASSO_{LF}$). As a comparison to these two signals, we also implement the adaptive LASSO proposed by Zou (2006), which is a statistical approach that balances the LASSO shrinkage with the standard regression estimates. Zou (2006) argues that the adaptive LASSO performs as well as if the true underlying model were known.

These alternative implementations of machine learning generate interesting results. The signal based on the adaptive LASSO method indeed has stronger predictive power than the simple LASSO for sector ETF returns. In regressions predicting next half-an-hour returns, the coefficient for the adaptive LASSO is on average 25% higher than that for the LASSO. Interestingly, the signals taking structural information into account are even more informative. Their regression slope coefficients in the predictive regressions are about twice that of the LASSO. When we form quintile portfolios based on the overnight signals, the average monthly return spread between Quintile 5 and Quintile 1 increases from 0.61% and 0.68% for the LASSO and adaptive LASSO, to 0.75%, and 0.69% based on the two methods using structural information. These results provide further support to the idea that economic links among firms provide an important mechanism for the rise of sectoral shocks. They also illustrate the advantages of structural machine learning.

Finally, we consider two composite signals that combine the individual predictors based

on different approaches. The first one simply takes an average of all predictors, while the second one combines signals via Gradient Boosted Regression Trees (GBRT). The results show that both composite signals can largely improve upon any of the individual LASSO predictors. In particular, the GBRT-signal exhibits the best performance, generating an equal-weighted average monthly return of 1.18% (t -statistic = 7.84) and a Sharpe ratio of 2.34, highlighting the advantage of boosted regression trees in combining signals.

Our results are robust to alternative empirical designs. First, although our main tests focus on high-frequency intraday data, the results also hold when we use daily and weekly data. Second, the results are robust when we use the net asset value instead of market prices to measure the sector ETF return, or when we use the Fama and French 49 industry portfolio returns over a longer sample period. Third, controlling for the intra-industry lead-lag effect as discovered by Hou (2007) does not influence our results. Fourth, instead of using the LASSO, we use the Elastic Net to train the signals, which generates similar results. Finally, to evaluate the investment value of our empirical findings, we construct a trading strategy that takes trading costs into account. The strategy **buys sector ETFs when the implied return based on the signals is greater than the historical average bid-ask spread and sells otherwise**. We find that depending on the specific trading signals, the monthly returns on the trading strategy after transaction cost range from 0.31% to 0.64%, with annualized Sharpe ratios between 0.59 and 1.30.

There is a growing literature on the microeconomic origins of macroeconomic shocks (Gabaix, 2011; Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi, 2012; Gabaix and Koijen, 2023; Galaasen, Jamilov, Juelsrud, and Rey, 2021). This literature emphasizes how shocks to microeconomic units generate aggregate impacts through concentration effects or propagation effects through economic links. Our study is motivated by insights from this literature. To the best of our knowledge, it is the first study that uses firm-level information scattered throughout the economy to predict sector-level fluctuations in the capital markets.

Our paper also contributes to the literature on industry return predictability. Most

studies in this literature use aggregate or sectoral information to explain industry return fluctuations (see, e.g., Boudoukh, Richardson, and Whitelaw, 1994; Moskowitz and Grinblatt, 1999; DellaVigna and Pollet, 2007; Bustamante and Donangelo, 2017; Rapach, Strauss, Tu, and Zhou, 2019), or use industry returns to forecast the aggregate economy (Hong, Torous, and Valkanov, 2007). What distinguishes our approach is that we exploit the granular information contained in the price movements of individual stocks to predict subsequent sector returns.

Due to our focus on high-frequency sector ETF returns, our paper adds to the literature on high-frequency return predictability in the stock market. Several studies examine the seasonality of high-frequency stock returns. For instance, Heston, Korajczyk, and Sadka (2010) report that stock returns tend to persist at the same half-hour intervals across trading days; Gao, Han, Li, and Zhou (2018) show that the first half-hour return on the market predicts the last half-hour return. Such seasonal return predictability is explained by the model of Bogousslavsky (2016) focusing on investors' infrequent portfolio rebalancing. In their study of intraday momentum across multiple asset classes, Baltussen, Da, Lammers, and Martens (2021) attribute the high-frequency return predictability to gamma hedging. Bogousslavsky (2021) explores anomaly returns during different times of the day. Chinco, Clark-Joseph, and Ye (2019) and Aït-Sahalia, Fan, Xue, and Zhou (2022) study return predictability at very short horizons. Chinco, Clark-Joseph, and Ye (2019) focus on stock return prediction at a one-minute frequency and find that stocks with news about fundamentals are likely to be identified as predictors by LASSO. Aït-Sahalia, Fan, Xue, and Zhou (2022) examine ultra short high-frequency return predictability ranging from milliseconds to a few minutes, and find that the predictability of returns vanishes quickly as the time span is larger than a few minutes. We contribute to this literature by providing novel evidence on the return predictability of sector ETFs at 30-minute and even lower frequencies. In addition, we make an initial step toward understanding the return predictability by studying the economic links among firms.

Finally, there is a rapidly growing literature that applies the techniques of machine learning to asset price anomalies and asset pricing models, as surveyed by Giglio, Kelly, and Xiu (2022). This literature shows the power of machine learning techniques in identifying the cross-sectional variation in individual stock returns (Freyberger, Neuhierl, and Weber, 2020; Gu, Kelly, and Xiu, 2020; Chen, Pelger, and Zhu, 2024; Cakici, Fieberg, Metko, and Zarembo, 2023), corporate bond returns (Bali, Goyal, Huang, Jiang, and Wen, 2022), mutual fund returns (Li and Rossi, 2021; DeMiguel, Gil-Bazo, Nogales, and Santos, 2023; Kaniel, Lin, Pelger, and Van Nieuwerburgh, 2023), and option returns (Bali, Beckmeyer, Moerke, and Weigert, 2023). Feng, He, and Polson (2019) apply deep learning to equity premium prediction. Feng, Giglio, and Xiu (2020) combine LASSO selection with Fama-MacBeth regressions to test new factor models by quantifying the gain of adding one new factor to a set of predefined factors. Jiang, Kelly, and Xiu (2023) apply the image-based convolutional neural networks (CNNs) trained using daily data to lower-frequency and international data for return prediction problems. Lopez-Lira and Roussanov (2022) challenge the central prediction of the arbitrage pricing theory using machine learning methods. Our paper contributes to the literature by showing the power of incorporating structural information into machine learning.

The remainder of the paper is organized as follows. Section 2 describes the data and empirical methodology. Section 3 presents the results using machine learning to aggregate firm-level information. Section 4 combines structural information with machine learning. Section 5 presents the results based on low-frequency data. Section 6 details a series of robustness checks. Section 7 concludes. The Internet Appendix details the high-frequency data cleaning rules and provides further evidence through simulation studies and additional robustness checks.

2. Empirical Methodology

2.1. Data Description

Our ETF sample consists of all sector ETFs from the ETFGlobal database. We obtain monthly holding data of the ETFs from the Center for Research in Security Prices (CRSP) Mutual Fund database. Our stock sample includes all firms listed on the New York Stock Exchange (NYSE), National Association of Securities Dealers Automated Quotations (NASDAQ), and American Stock Exchange (AMEX) with share code of 10 or 11. Our intraday price and quote data for ETFs and stocks come from the NYSE trade and quote (TAQ) database. The final sample includes 234 unique sector ETFs and 10,076 unique stocks over the period of 5,178 days between January 2000 and July 2020.² The average number of unique stocks per day is around 4,300.

To prepare the intraday return data, we collect minute-by-minute observations of intraday prices by applying the cleaning rules of Bollerslev, Li, and Todorov (2016), Bollerslev, Li, and Zhao (2020), and Jiang, Li, and Wang (2021) to the TAQ database. Our high-frequency data cleaning procedures are detailed in the Internet Appendix. Based on intraday prices, we then compute intraday returns as every 30-minute return between 10:00 and 16:00 and the overnight return as the return between 16:00 on the previous trading day and 10:00 on the current trading day.³ Since transaction prices from TAQ are raw prices without adjusting for corporate actions such as dividend payout and stock splits, we apply the daily “cumulative factor to adjust price” and “dividend cash amount” variables in the CRSP database to adjust for split and dividend.

The high-frequency firm-level news data are from the RavenPack news database, which

²Fig. A.1 in the Internet Appendix plots the snapshot of the percentage sector ETFs in each industry by the end of our sample period in July 2020. Industries are classified based on the IO industry code from the BEA; only the top 16 industries with at least 1% ETFs are displayed. Among these industries, the one with the greatest number of ETFs is electronic instruments (18.5%) and the one with the least number of ETFs is apparel (1.23%).

³We use the price at 10:00 for overnight returns to ensure that most securities have traded at least once after the market open.

provides a comprehensive sample of firm-specific news stories from the Dow Jones News Wire (Recent studies using this data set include Kelley and Tetlock, 2017; Jiang, Li, and Wang, 2021). To ensure a news story is specifically about a given firm, we rely on the “relevance score” variable provided by RavenPack, which captures how closely the news story is related to a particular company. The score ranges from 0 to 100, where a score of 0 (100) means that the entity is passively (predominantly) mentioned. We only use news stories with a relevance score of 100 in our sample. To include only news stories about firm fundamentals, we select 12 news groups of acquisitions-mergers, analyst-ratings, assets, bankruptcy, credit, credit-ratings, dividends, earnings, equity-actions, labor-issues, product-services, and revenues from a total of 29 news groups. To keep only fresh news about a company, we further exclude repeated news by requiring the “event novelty score” from RavenPack to be 100. Applying these filters does not introduce look-ahead bias because all news articles are processed by RavenPack within milliseconds of receipt and the resulting data are immediately sent to subscribers. Thus, the information is available in real time.

2.2. *The Machine-Learning Predictors*

2.2.1. *LASSO*

Our first machine-learning predictor is the LASSO signal that aggregates high-frequency price movements of a broad universe of individual stock returns. Given an input of the high-frequency returns of p individual stocks $(r_{1,d,i}^{stc}, r_{2,d,i}^{stc}, \dots, r_{p,d,i}^{stc})$ on day d in interval i , we wish to predict the j th sector ETF return $r_{j,d,i+1}^{etf}$ on day d in interval $i+1$. In this paper, we focus on intraday frequency of 30 minutes and consider the overnight period as from 16:00 to 10:00. Therefore, we have $i = 1, 2, \dots, 13$ with $i = 1$ indicating the overnight period and $i = 2$ the period between 10:00 and 10:30 and so on until $i = 13$ indicating the interval between 15:30 and 16:00. Thus, for each high-frequency interval, we propose the following

linear model:

$$r_{j,d,i+1}^{etf} = \beta_{j,i,0} + \sum_{s=1}^p \beta_{j,i,s} r_{s,d,i}^{stc} + \epsilon_{j,d,i+1}. \quad (1)$$

Note that this specification allows each ETF to have its specific regression coefficients for a given high-frequency interval, to gain the flexibility in modeling the sector ETFs.

To estimate the regression coefficients, we have two considerations. First, for each sector ETF, we do not expect all the stocks to be useful in predicting its return. Instead, we expect that a subgroup of stocks may be particularly informative in predicting the ETF return. Second, the input space is relatively large (thousands of stocks) in comparison with the sample size (thousands of observations). In addition, the return prediction problem is in a world of low signal-to-noise ratio. Under such situations, statistical machine learning theory tells us that we can improve the prediction accuracy by introducing constraints to reduce the estimation variance at the cost of sacrificing a smaller amount of estimation bias. Motivated by these two considerations, we use the **Least Absolute Shrinkage and Selection Operator** (LASSO) to estimate these coefficients. LASSO can sparsely estimate the coefficients $\{\beta_{j,i,s}\}$ by automatically setting many of them to be exactly zero, thus achieving both stock selection and variance reduction goals arising from our two considerations.

LASSO encourages sparse estimates of regression coefficients by introducing the L_1 penalty to the standard least squares for regressions. Let $\beta_{j,i} = (\beta_{j,i,0}, \beta_{j,i,1}, \dots, \beta_{j,i,p})$ be the regression coefficients of predicting the j th ETF in the i th high-frequency interval. Using the LASSO penalty, we will minimize the following loss function:

$$\hat{\beta}_{j,i}^{LASSO} = \underset{\beta_{j,i}}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_d \left(r_{j,d,i+1}^{etf} - \beta_{j,i,0} - \sum_{s=1}^p \beta_{j,i,s} r_{s,d,i}^{stc} \right)^2 + \lambda_{j,i} \sum_{s=1}^p |\beta_{j,i,s}| \right\}, \quad (2)$$

where λ is the hyperparameter that controls the amount of shrinkage. Large values of λ induce more zeros in the corresponding $\hat{\beta}_{j,i}^{LASSO}$. In the extreme case of $\lambda = 0$, there will be no penalty so $\hat{\beta}_{j,i}^{LASSO}$ corresponds to the ordinary least squares estimates. In the other

extreme case of $\lambda = \infty$, the objective function will be dominated by the penalty term and thus $\hat{\beta}_{j,i}^{LASSO} = 0$. Thus, the choice of λ is key to the performance of LASSO. In machine learning, the most common method for hyperparameter tuning is the validation set approach. The basic idea is to use only part of the data for fitting and a different part for validation. Then we will choose the λ that has the smallest prediction error in the validation set for generating LASSO signals. Intuitively, this process that fits the model on the training sample and then evaluates the fit on the validation sample mimics the process of in-sample training and then out-of-sample prediction. Therefore the λ value that works better in validation set tends to perform better out-of-sample than other values of λ .

2.2.2. Adaptive LASSO

The LASSO tends to underestimate the magnitude of non-zero coefficients in sparse settings because all coefficients are shrunk at a constant rate controlled by a single λ which tends to be large in sparse models. One approach for reducing such bias is the adaptive LASSO by Zou (2006), which modifies the LASSO penalty function so that larger coefficients are shrunk less severely. More formally, the adaptive LASSO estimator is given by:

$$\hat{\beta}_{j,i}^{LASSO_{ADP}} = \underset{\beta_{j,i}}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_d \left(r_{j,d,i+1}^{etf} - \beta_{j,i,0} - \sum_{s=1}^p \beta_{j,i,s} r_{s,d,i}^{stc} \right)^2 + \lambda_{j,i} \sum_{s=1}^p \hat{w}_{j,i,s} |\beta_{j,i,s}| \right\}, \quad (3)$$

where the weight is in the form of $\hat{w}_{j,i,s} = 1/|\hat{\beta}_{j,i,s}^{OLS}|^v$, $\hat{\beta}_{j,i,s}^{OLS}$ is the ordinary least squares (OLS) estimator, and $v > 0$. For simplicity and ease of interpretation, we set $v = 1$ so that the weights are given by:

$$\hat{w}_{j,i,s} = \frac{1}{|\hat{\beta}_{j,i,s}^{OLS}|}. \quad (4)$$

This approach is known as a two-stage approach. In the first stage, we run OLS regressions to generate the $\hat{\beta}_{j,i}^{OLS}$ estimates which in turn determine the weight vector $\hat{\mathbf{w}}_{j,i}$. In the second stage, we apply the LASSO under the modified loss function in Eq. (3). Zou (2006) proves

that the adaptive LASSO has oracle properties, namely, it performs as well as if the correct submodel were known in advance (Fan and Li, 2001).

2.2.3. *Customer-Supplier-Based LASSO and Leading-Firm-Based LASSO*

Another approach to increase the power of LASSO estimates is through feature screening, where we reduce the high dimensionality of the feature space (e.g., thousands of stocks) to a moderate size (e.g., a few hundred or a few dozens of stocks) before applying sophisticated LASSO procedures. To screen features, standard methods often rely on statistical models such as correlation-based approaches. For our sector ETF return prediction problem, we propose two new screening methods motivated by economics. The first utilizes the customer-supplier relation and the second uses the leading firms based on analyst coverage.

To screen predictors, one possible way is to rely on the economic relationship between our target (sector ETFs) and predictors (individual stocks). Cohen and Frazzini (2008) find evidence of return predictability across economically linked firms in the presence of investors' attention constraints. Menzly and Ozbas (2010) further document that stocks that are in economically related supplier and customer industries can cross-predict each other's returns due to investor specialization and market segmentation.⁴ Therefore, for each sector ETF j , we select economically linked firms as the only predictors by setting $\hat{w}_{j,i,s}$ in Eq. (3) as:

$$\hat{w}_{j,i,s}^{CS} = \begin{cases} 1 & \text{if } s \in \Theta_j, \\ \infty & \text{if } s \notin \Theta_j, \end{cases} \quad (5)$$

where Θ_j indicates a set of firms that are economically linked to ETF j . This is equivalent to including only economically linked firms in Eq. (2), or minimizing the following loss function:

⁴On a related note, Jang, Wang, and Zhang (2021) show that firms' international activities can deliver relevant information to investors and predict stock returns.

$$\hat{\beta}_{j,i}^{LASSOCS} = \underset{\beta_{j,i}}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_d \left(r_{j,d,i+1}^{etf} - \beta_{j,i,0} - \sum_{s=1}^p \beta_{j,i,s} r_{s,d,i}^{stc} \right)^2 + \lambda_{j,i} \sum_{s=1}^p \hat{w}_{j,i,s}^{CS} |\beta_{j,i,s}| \right\}. \quad (6)$$

To select economically linked firms, we rely on the 2002 Input-Output (IO) Surveys of the Bureau of Economic Analysis (BEA). The BEA IO surveys provide data on the amount of goods and services traded among industries, and thus allow us to identify supplier and customer industries. The BEA surveys report trades in two tables – the Make and Use Tables. The Make Table reports the value of each commodity produced by each industry, and the Use Table reports the value of each commodity purchased by each industry. Because each commodity is mainly produced by one industry (i.e., a nearly one-to-one mapping between commodities and industries in the Make Table), we follow Menzly and Ozbas (2010) and solely rely on the Use Table to identify customer-supplier (C-S) links.

The BEA uses a single set of IO industry codes to classify both industries and commodities. We first assign each stock an industry based on its reported North American Industry Classification System (NAICS) code from the COMPUSTAT, and then use the mapping between NAICS and IO industry codes provided by the BEA to assign an IO industry code to each stock. Following Menzly and Ozbas (2010), we merge three separate industry accounts, namely, 2301 (new nonresidential construction), 2302 (new residential construction), and 2303 (maintenance and repair construction) into a single account because these three industry accounts share overlapping claims on the same NAICS codes and firms, and also drop miscellaneous industry accounts related to government, import, and inventory adjustments because such accounts cannot be clearly mapped to standard industries.

To assign an IO code to an ETF, we assume that the industry each ETF belongs to is determined by the majority of the ETF's constituent firms. For example, if most of ETF j 's constituent firms belong to the "Food products" industry, we assign the IO code of "Food products" to ETF j . Next, we find all industries that have a C-S relationship

with the industry of the target ETF j (e.g., “Food products”) based on the consumption value of the Use Table. For instance, the most related industries to “Food products” are “Food manufacturing” and “Food services and drinking places”, whereas the least related industries include “Leather and allied product manufacturing” and “Internet publishing and broadcasting”. To focus on the most economic-linked firms (predictors) with the target ETF, we select industries that have an above-median consumption value. Finally, we collect all unique firms from the selected industries, resulting in an average of 577 C-S linked firms (predictors) for each target ETF.

An alternative way of screening predictors is to identify firms that lead returns of other stocks. Hameed, Morck, Shen, and Yeung (2015) show that stocks disproportionately followed by analysts can best predict returns of their industry peers. We follow Hameed, Morck, Shen, and Yeung (2015) and define leading firms based on analyst coverage from the Institutional Brokers’ Estimate System (IBES) database. To be consistent with our identification of C-S linked firms, we first group stocks into different industries by using the IO industry codes from the BEA. Then, for each firm within the same industry, we define those that have an above-median number of analysts as leading firms and set $\hat{w}_{j,i,s}$ in Eq. (3) as:

$$\hat{w}_{j,i,s}^{LF} = \begin{cases} 1 & \text{if } s \in \Lambda_j, \\ \infty & \text{if } s \notin \Lambda_j, \end{cases} \quad (7)$$

where Λ_j indicates a set of all leading firms that are in the same industry as the target ETF j . The loss function becomes:

$$\hat{\beta}_{j,i}^{LASSO_{LF}} = \underset{\beta_{j,i}}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_d \left(r_{j,d,i+1}^{etf} - \beta_{j,i,0} - \sum_{s=1}^p \beta_{j,i,s} r_{s,d,i}^{stc} \right)^2 + \lambda_{j,i} \sum_{s=1}^p \hat{w}_{j,i,s}^{LF} |\beta_{j,i,s}| \right\}. \quad (8)$$

Equivalently, we include only leading firms in Eq. (2). On average, there are 95 leading firms (predictors) for each target ETF.

2.2.4. Implementation

We implement the various LASSO methods in a rolling-fit fashion, where each fit consists of three steps: training, validation and testing. In particular, at the end of each year y , we use the lagged returns in interval i of the entire cross section of stocks that are available for the five-year period between year $y-4$ and y as predictors.

Training: We use four-year data between years $y-4$ and $y-1$ to train model L in Eq. (2), (3), (6), or (8) on an *ETF-by-ETF* and *interval-by-interval* basis for a set of candidate hyperparameters λ covering a wide range of sparsity level (e.g., separate models are trained for each sector ETF at each high-frequency interval).

Validation: Using the estimated coefficients from training, we generate the return predictions for each ETF for the validation sample consisting of the data in year y . Then based on these predictions for year y and their corresponding true ETF returns, we compute the mean squared errors as the evaluation metric and estimate the optimal $\hat{\lambda}_{j,i}^L$ to be the one with the smallest mean squared errors.

Testing: Finally, we use the coefficients $\hat{\beta}_{j,i}^L$ estimated under $\hat{\lambda}_{j,i}^L$ for generating the various LASSO signals for year $y+1$:

$$f_{j,d,i}^{etf,L} = \hat{\beta}_{j,i,0}^L + \sum_{s=1}^p \hat{\beta}_{j,s,i}^L r_{s,d,i}^{stc}, \quad (9)$$

where $L \in (LASSO, LASSO_{ADP}, LASSO_{CS}, LASSO_{LF})$. The notation $f_{j,d,i}^{etf,L}$ stands for the signal for the j th ETF on day d at the end of interval i using LASSO method L . We repeat the above steps for each year from 2004 to 2019.

2.2.5. Firms Selected by Different LASSO Algorithms

Before evaluating the predictive performance of our signals, a natural question is: how many stocks are selected by the LASSO algorithms to predict sector ETFs? Fig. 1 plots the mean and 90% confidence interval for the number of firms selected by various LASSO

methods to generate the overnight LASSO signals for predicting the first half-an-hour return each year during our sample period.⁵ Panel A displays the result of the *LASSO* method. There are several noteworthy patterns. First, on average, LASSO selects only 35 predictors (out of around 4,300 possible predictors) when making its first-30-minute return forecasts for each ETF. This finding shows that the stock signals for sector ETFs are indeed sparse. The use of the sparsity-encouraging techniques such as LASSO is critical to the discovery of these sparse signals. Theoretically, Donoho (2006) shows that under certain assumptions on the model matrix, if the true model is sparse, the LASSO solution identifies the correct predictors with high probability. Second, LASSO tends to select more predictors during the financial crisis. For example, LASSO uses more than 100 predictors for almost all ETFs in 2008. This is somehow consistent with the findings that correlations are much elevated during extreme downside periods (see, e.g., Ang and Chen, 2002).

Panels B, C and D plot the time variation of the number of firms selected by *LASSO_{ADP}*, *LASSO_{CS}*, and *LASSO_{LF}*, respectively. All three panels exhibit similar patterns as Panel A. For example, the number of selected firms is small on average (10 stocks) and spikes during the 2008 financial crisis for each of the two-stage LASSO methods. However, the last three panels also demonstrate some distinct features different from Panel A. The most notable difference is that the number of selected firms is relatively stable in the cross section as evident from the narrower confidence intervals. Another difference is that, on average, the two-stage LASSO methods select a smaller number of firms than the simple LASSO.

Fig. 2 plots the mean and 90% confidence interval for the proportion of ETF constituent firms selected by various LASSO algorithms.⁶ Panels A and B show that, on average, the average proportion of constituent firms is only 3.7% for the simple *LASSO* signal and 1.1% for the *LASSO_{ADP}* signal, respectively. The results imply that most part of the industry

⁵The number of predictors selected by LASSO is controlled by the hyperparameter λ chosen by validation. We use fixed 1-year data as our validation set. The first (last) validation set in our sample period is in 2004 (2019).

⁶To calculate the proportion of constituent firms, the number of overlapping firms between LASSO-selected and ETF constituent firms is normalized by the total number of ETF constituents for the *LASSO* and *LASSO_{ADP}* signals. For *LASSO_{CS}* (*LASSO_{LF}*), it is normalized by the number of overlapping firms between C-S linked (leading) firms and ETF constituent firms.

return predictability based on the pure statistical LASSO and adaptive LASSO methods may come from firms other than the constituents. Put it differently, these machine learning methods can be extremely useful in identifying non-constituent firms as predictors. However, as seen from Panels C and D, adding economic constraints to the LASSO weights can result in a relatively higher proportion of constituent firms (that are also subsets of the screened firms) selected by LASSO. For instance, on average, $LASSO_{CS}$ selects 10.2% constituent firms, and the ratio jumps to 27.5% for $LASSO_{LF}$. Thus, while pure machine learning methods such as LASSO and adaptive LASSO select predictors solely from a statistical perspective, adding human intervention (e.g., $LASSO_{CS}$ and $LASSO_{LF}$) may help machine select predictors from an economic point of view.

Another intriguing question is, which kind of firms in the entire stock universe are more likely to be selected by the simple LASSO algorithm to predict sector ETF returns? If there exists within-industry lead-lag effects or lead-lag effects along the supply chain, two types of firms will be more likely selected as predictors: 1) firms in the same industry of the sector ETF (i.e., within-industry firms), and 2) firms in the C-S linked industries of the sector ETF (i.e., C-S linked firms). To assess the likelihood of these firms to be selected by LASSO, for each sector ETF and each LASSO training sample, we calculate the ratio of within-industry firms among all LASSO-selected firms and the ratio of C-S linked firms among all LASSO-selected firms. Then, for each ETF, we compute the average of these two ratios over training samples. Lastly, we plot the histogram of the average ratios among ETFs in Fig. 3 (ratio for within-industry firms in Panel A and that for C-S linked firms in Panel B). Interestingly, LASSO tends to select fewer within-industry firms but more C-S linked firms. For example, for nearly 70% sector ETFs, the average ratio of within-industry firms is below 10%, whereas the average ratio of C-S linked firms is above 90%.

2.3. The News-Driven Return Predictor

Our news signal is a fundamental-driven, news-based predictor that aggregates fundamental information of individual stocks for predicting sector ETF returns. The signal is constructed via a bottom-up process.

Stock-Level News-Driven Returns: First, we construct stock-level news-driven returns. Following Jiang, Li, and Wang (2021), we classify high-frequency stock returns $r_{s,d,i}^{stc}$ as news-driven returns based on high-frequency market reaction to information. Specifically, if at least one firm-level news is released during the return measurement period, the news-driven return is equal to the total return of that period; the news-driven return is set to zero otherwise. For news occurring within regular trading hours, the news return is simply the 30-minute return over the same period that the news occurs. For news occurring outside regular trading hours such as during the weekend, holiday, or overnight, the news return is the surrounding overnight return to reflect the fact that such news stories are incorporated into the stock's prices by the first few trades on the following trading day. For example, the return for news events during the weekend is the return over the period of 16:00 of the surrounding Friday and 10:00 of the surrounding Monday.

ETF-Level News-Driven Returns: After obtaining the high-frequency firm-level news returns, we construct high-frequency ETF-level news returns as the value-weighted average of news returns of its constituent firms. More formally, suppose there are N constituent firms in a given ETF j on day $d-1$, and let $r_{s,d,i}^{stc,news}$ denote the news return of constituent firm s on day d in interval i . The high-frequency news-driven ETF signal on day d and by the end of interval i is computed as:

$$f_{j,d,i}^{etf,news} = C_{j,d,i} \sum_{s=1}^N w_{j,d-1,s} r_{s,d,i}^{stc,news}, \quad (10)$$

where $w_{j,d-1,s}$ is the weight of constituent firm s in ETF j on day $d-1$; ⁷ $C_{j,d,i}$ is a normalizing

⁷The weights of ETFs on constituent firms are based on monthly holding data from CRSP. For each trading day d , we rely on holdings from the nearest previous day to approximate the weights on day $d-1$.

constant such that the total weights on constituent firms with non-zero new return is equal to one (i.e., $C_{j,d,i} \sum_{s=1}^N w_{j,d-1,s} \mathbf{1}(r_{s,d,i}^{stc,news} \neq 0) = 1$). If all constituent firms have zero news return on day d in interval i , $r_{j,d,i}^{etf,news}$ is set to be zero.

2.4. Control Variables

In addition to the three proposed signals, we construct a number of control variables at ETF level following standard definitions in the literature. *RET* is the lagged overnight or 30-minute return. *ILLQ* is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day (Amihud, 2002). *RVOL* is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. *Mom* for a given day t is the cumulative ETF return from day $d-252$ to day $d-21$. *Size* is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily.

2.5. Descriptive Statistics

Since corporate news tends to arrive disproportionately after market close, we start with the evaluation of the overnight signals (i.e., $i = 1$ in Eqs. (9) and (10)) that are generated at the market open of each day. Table 1 reports descriptive statistics of the overnight *LASSO*, *LASSO_{ADP}*, *LASSO_{CS}*, *LASSO_{LF}*, and *News* signals, as well as other control variables for all ETFs in our sample. All signals and the lagged return *RET* are reported in basis points (bps). Panel A shows the mean, standard deviation, and five quantiles, all of which are first computed cross-sectionally on each day and then averaged over time. The results imply that the distribution of the four overnight LASSO signals are similar to each other and all close to symmetric. The mean and median of the LASSO signals are around -0.3 and -0.2 bps, respectively. The distribution of the overnight news signal is slightly skewed to the

right with a mean of 4.03 bps and a median of 4.96 bps. The cross-sectional dispersion of the overnight news signal as measured by their cross-sectional standard deviation is 165.07 bps, which is greater than the dispersion of the LASSO signals at around 8 bps. Among the four LASSO signals, $LASSO_{LF}$ has the smallest dispersion of 6.47 bps, while $LASSO$ has the largest dispersion of 8.96 bps.

Panel B reports the average cross-sectional correlations among these variables. The four LASSO signals exhibit relatively low pairwise correlation, with the highest correlation of 0.538 between $LASSO_{CS}$ and $LASSO_{LF}$ and the lowest correlation of 0.241 between $LASSO$ and $LASSO_{LF}$. Therefore, different weighting schemes in training LASSO can result in quite different estimators. Furthermore, all LASSO signals are uncorrelated with the news signal, with a correlation coefficient of around 0.02. The correlations between our LASSO signals and the control variables are also low, indicating the uniqueness of our signals in comparison with existing ones. The overnight news signal is only highly correlated with the overnight return with a correlation of 0.44, which could be driven by some large price jumps accompanying news (Jiang, Li, and Wang, 2021). On the other hand, the correlation structure among the control variables is consistent with previous literature. For instance, we find that ETFs with high net asset value tend to have higher liquidity and lower volatility.

3. Using Machine Learning to Aggregate Firm-Level Information

3.1. Intraday Half-Hour Prediction

We start with the evaluation of the performance of the simple $LASSO$ and news signals for predicting intraday 30-minute returns. First, we generate the signal values for every interval. Specifically, we follow the same procedure as in Eqs. (9) and (10) to generate high-frequency signals $f_{j,d,i-1}^{etf,LASSO}$ and $f_{j,d,i-1}^{etf,news}$ that will be used to forecast return in 30-minute interval i between 10:00 and 16:00 on day d .

After obtaining the high-frequency signals, we run the Fama and MacBeth (1973) regressions to test their predictive power for the subsequent 30-minute ETF returns. Specifically, for each 30-minute return in interval $i = 2, 3, \dots, 13$ on day d , we perform the following cross-sectional regressions:

$$r_{j,d,i}^{etf} = \gamma_{0,d,i} + \gamma_{LASSO,d,i} f_{j,d,i-1}^{etf,LASSO} + \gamma_{news,d,i} f_{j,d,i-1}^{etf,news} + \sum_{k=1}^K \gamma_{k,d,i} Z_{j,d-1,k} + \epsilon_{j,d,i}, \quad (11)$$

where the control variables $Z_{j,d-1,k}$ for ETF j are all measured at the end of day $d - 1$. For a given 30-minute forecasting interval, we obtain on each day the slope coefficients from these cross-sectional regressions. We then compute the time-series average of each slope coefficient to test if the signals are statistically significant in predicting forward returns. Our control variables include the lagged overnight or 30-minute return, the illiquidity measure of Amihud (2002), realized volatility, ETF returns from day $d-252$ to day $d-21$ as a proxy for price momentum, and ETF size.

Table 2 reports the estimated average slope coefficients and corresponding t -statistics for predicting returns in different 30-minute intervals of the day. The results clearly show that the LASSO signals have positive and significant predictive power for every 30-minute interval, with t -statistics above five across all intervals. For the news signal, we find that its overnight version can significantly predict the first half-an-hour return with a t -statistic above 21. The coefficients of the news signal in other intervals are all insignificant except for the 14:30 to 15:00 interval. This is because news stories are less frequently released during trading hours. As shown in Panel A of Fig. 4, on average, 180 constituent firms have news after market close, whereas less than 15 constituent firms have news within each 30-minute interval during market open. Panel B in Fig. 4 plots the intraday distribution of the number of ETFs with news. It shows that, on average, around 100 ETFs have constituents with news before 10:00 market open. However, only around 20 ETFs have news constituents within each 30-minute interval after market open. As a result, the per interval Fama-MacBeth regression might not have enough statistical power for testing the significance of the news signal.

3.2. Longer Horizon Prediction

Our prediction evaluation analyses have been focusing on the signals computed at each high-frequency interval. We find that while each overnight or intraday *LASSO* signal can predict the next half-an-hour return, the news signal only has strong predictive power for its overnight version. We now dig more into the performance of the overnight signals that are generated at the market open of each day. The market open period also tends to be the most liquid during the entire trading session. To evaluate the predictive performance over different horizons, we construct different forward returns all starting from the same market open at 10:00 but end at different times of the day. Let $r_{j,d,2:i}^{etf}$ be the return between the start of interval 2 (e.g., 10:00) to the end of interval i . That is, for $i = 2, 3, \dots, 13$, $r_{j,d,2:i}^{etf}$ are the returns between 10:00 and 10:30, 10:00 and 11:00, and so on until between 10:00 and 16:00. We then evaluate the predictive power of the two overnight signals $f_{j,d,1}^{etf,LASSO}$ and $f_{j,d,1}^{etf,news}$ in Eqs. (9) and (10).

We continue to use the Fama and MacBeth (1973) regressions to test the predictive power of the signals by performing the following cross-sectional regressions:

$$r_{j,d,2:i}^{etf} = \gamma_{0,d,2:i} + \gamma_{LASSO,d,2:i} f_{j,d,1}^{etf,LASSO} + \gamma_{news,d,2:i} f_{j,d,1}^{etf,news} + \sum_{k=1}^K \gamma_{k,d,2:i} Z_{j,d-1,k} + \epsilon_{j,d,2:i}. \quad (12)$$

Table 3 reports the estimated average slope coefficients and their t -statistics for predicting different forward returns. The case with $i = 2$ (i.e., predicting the 30-minute ahead return between 10:00 and 10:30) in the first column is the same as the first column in Table 2 and is included for easy comparison. The results clearly indicate that both the overnight *LASSO* and news signals positively predict longer intraday ETF returns. For the *LASSO* signal, the slope coefficients range from 0.101 with a t -statistic of 6.28 when predicting 30-minute ahead returns between 10:00 and 10:30 to 0.148 with a t -statistic of 4.68 when predicting 300-minute ahead returns between 10:00 and 15:00. For the news signal, the coefficients vary

from 0.021 with a t -statistic of 21.06 when predicting 30-minute ahead returns to 0.028 with a t -statistic of 14.93 when predicting open-to-close returns. The coefficients of the LASSO and news signals also tend to increase along with the forecasting horizon, indicating the predictive power of both signals could last through market close.

To put these regression coefficients into perspective, the average cross-sectional standard deviation of the overnight LASSO signal from Table 1 equals 8.96 bps. Thus, when predicting open-to-close returns, the coefficient of 0.121 for the LASSO signal implies that a two-standard-deviation increase in the LASSO signal predicts a rise of approximately 5.5% in annual ETF returns ($2 \times 8.96 \times 10^{-4} \times 0.121 \times 252 = 5.5\%$). Similarly, the coefficient of 0.028 for the news signal indicates that a two-standard-deviation increase in the news signal predicts a rise of around 23.3% per annum in future ETF returns ($2 \times 165.07 \times 10^{-4} \times 0.028 \times 252 = 23.3\%$). Hence, the predictive powers of the overnight LASSO and news signals are both statistically and economically significant.

Can the two overnight signals predict returns only until the same-day market close or much longer? To answer this question, we re-fit Eq. (12) by replacing the dependent variable with interdaily returns from 10:00 on day d to 16:00 on day $d+k$ for each $k = 0, 1, \dots, 4$, and present the coefficients and t -statistics in Table 4. The case with $k = 0$ in the first column is the same as the second-to-last column in Table 3 and is included for comparison. As the forecast horizon increases, the predictive power of the overnight LASSO signal remains statistically significant. However, its strength decreases as is evident from the smaller t -statistics of 1.82 for $k = 4$ in comparison with the t -statistics of 3.79 for $k = 0$. On the other hand, the overnight news signal remains a powerful predictor at longer horizons. For example, its slope coefficient increases from 0.028 when predicting one-day open-to-close return to 0.032 when predicting five-day returns, and the t -statistics are above five for all horizons, consistent with the long-lasting news momentum effect documented by Jiang, Li, and Wang (2021). In summary, the Fama-MacBeth regressions support the predictive power of the LASSO and news signals.

4. Incorporating Structural Information into Machine Learning

4.1. Intraday Half-Hour Prediction

In this section, we present the results of the other three two-stage LASSO methods defined in Section 2.2, namely $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$.

We start with evaluating the performance of the intraday LASSO signals for predicting 30-minute returns. Specifically, for each 30-minute interval between 10:00 and 16:00 (i.e., $i = 2, \dots, 13$), we perform the following cross-sectional regressions:

$$r_{j,d,i}^{etf} = \gamma_{0,d,i} + \gamma_{LASSO,d,i} f_{j,d,i-1}^{etf,L} + \gamma_{news,d,i} f_{j,d,i-1}^{etf,news} + \sum_{k=1}^K \gamma_{k,d,i} Z_{j,d-1,k} + \epsilon_{j,d,i}, \quad (13)$$

where $f_{j,d,i-1}^{etf,L}$ is defined by Eq. (9) and $L \in (LASSO_{ADP}, LASSO_{CS}, LASSO_{LF})$.

Panels A to C in Table 5 report the estimated average slope coefficients and the corresponding t -statistics for predicting 30-minute ahead returns using $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$, respectively. The cross-interval average adjusted R^2 increases from 14.51% to 16.23% when we switch from $LASSO$ to $LASSO_{ADP}$, indicating that the adaptive LASSO can improve the predictive power of the LASSO by using the weighted penalty to reduce the bias in the initial LASSO estimates. Different from $LASSO$ and $LASSO_{ADP}$, the $LASSO_{CS}$ signal presented in Panel B is constructed from choosing firms that are economically linked to the target ETF as the predictors. Interestingly, $LASSO_{CS}$ can further improve the predictive power of $LASSO_{ADP}$ across most intervals of the day with a cross-interval average adjusted R^2 of 16.53%. Panel C presents the results for $LASSO_{LF}$ constructed based on leading firms in the industry of the target ETF. As can be seen, $LASSO_{LF}$ yields the best performance among all LASSO signals with an average adjusted R^2 as high as 16.78%. The stronger predictive power of $LASSO_{CS}$ and $LASSO_{LF}$ over $LASSO$ and $LASSO_{ADP}$ further demonstrates the importance of human intervention and feature engineering based on priori

knowledge of firm connection in improving machine learning performance.

4.2. Longer Horizon Prediction

The previous section shows that the three two-stage LASSO signals can improve over the simple *LASSO* signal in predicting 30-minute ahead returns. Would their relative performance remain superior at longer return forecasting horizons? To answer this question, we now rely on various overnight LASSO signals to predict longer intraday or interdaily returns.

We first evaluate the predictive performance of the three alternative overnight LASSO signals over various intraday horizons from 10:00 market open to different times of the day. Specifically, we run the following Fama and MacBeth (1973) cross-sectional regressions for each alternative LASSO signal:

$$r_{j,d,2:i}^{etf} = \gamma_{0,d,2:i} + \gamma_{f,d,2:i} f_{j,d,1}^{etf,L} + \gamma_{news,d,2:i} f_{j,d,1}^{etf,news} + \sum_{k=1}^K \gamma_{k,d,2:i} Z_{j,d-1,k} + \epsilon_{j,d,2:i}, \quad (14)$$

where $f_{j,d,i-1}^{etf,L}$ is defined by Eq. (9), $L \in (LASSO_{ADP}, LASSO_{CS}, LASSO_{LF})$, and $r_{j,d,2:i}^{etf}$ stands for the return between the start of interval 2 (e.g., 10:00) and the end of interval i . That is, for $i = 2, 3, \dots, 13$, $r_{j,d,2:i}^{etf}$'s are the returns between 10:00 and 10:30, 10:00 and 11:00, and so on until between 10:00 and 16:00.

Panels A to C in Table 6 summarize the average estimated slope coefficients and the t -statistics for predicting different forward returns using $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$, respectively. Overall, the three two-stage LASSO signals all yield stronger predictive power than the *LASSO* signal in Table 3 in terms of higher average signal coefficient and adjusted R^2 's across horizons. For instance, the cross-horizon average signal coefficients for $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ are around 0.23, 0.29, and 0.35 compared to the average coefficient of 0.12 for *LASSO*, and the average adjusted R^2 's for the three signals are 19.51%, 19.70%, and 19.89% compared to the average R^2 of 18.03% for *LASSO*. Similar to the *LASSO* signal, the coefficients of the three two-stage signals also

tend to increase along with the forecasting horizon, indicating persistent predictive power until market close. To gauge the economic significance, we focus on predicting open-to-close ETF returns using each overnight LASSO signal. A two-standard-deviation increase in the $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signal predicts a rise of annual ETF return around 7.31%, 10.43%, and 11.09%, respectively, all dominating the rise of annual ETF return around 5.5% based on $LASSO$. The stronger predictive power of the three two-stage signals, especially $LASSO_{CS}$ and $LASSO_{LF}$, further highlights the importance of screening predictors based on economic connections.

We next investigate the predictive power of the alternative LASSO signals over horizons longer than one day by re-fitting Eq. (14). Specifically, we replace the dependent variable with interdaily returns from 10:00 on day d to 16:00 on day $d+k$ for each $k = 0, 1, \dots, 4$, and present the coefficients and t -statistics in Table 7. The predictive power of the three signals remains statistically significant for up to five days and all dominates that of the simple $LASSO$ signal as is evident by the greater signal coefficients and adjusted R^2 across the board. Among the three signals, $LASSO_{LF}$ has the strongest predictive power across horizons with an average coefficient around 0.42, more than twice the magnitude of the average $LASSO$ signal coefficient at 0.15 from Table 4. The predictive power of $LASSO_{CS}$, albeit statistically significant, somewhat decreases as the forecasting horizon increases. One possible explanation of the contrast is that the information from industry leaders is slowly incorporated into the return of the followers, generating long-lasting ETF predictability, whereas the information along customer-supplier chain is transmitted faster.

4.3. Portfolio Sorts

So far, we have been focusing on the regression analysis of the LASSO signal performance. We next form portfolios based on the overnight version of the four LASSO signals. Specifically, at 10:00 of each trading day d , we sort ETFs into quintile portfolios based on their overnight

$LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals.⁸ Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom quintile with low signal values, with a holding period from 10:00 to 16:00 of the same trading day.

Table 8 summarizes the monthly portfolio returns in percentage, which are converted from daily returns by multiplication with a factor of 21 for ease of interpretation. The row labeled “High-Low” reports the average return spread between Quintiles 5 and 1, and the row labeled “CAPM Alpha” reports return spread adjusted by market exposure. Consistent with the results of regression analysis, Table 8 demonstrates an increasing relation between signal values and future ETF returns. For instance, the average monthly returns increase from 0.04% in Quintile 1 with low overnight $LASSO$ signal to 0.65% in Quintile 5 with high overnight $LASSO$ signal, yielding a return spread of 0.61% per month with a t -statistic of 4.52. The quintile returns sorted by the other three two-stage $LASSO$ signals show a similar pattern with more significant return spreads. Specifically, the high-minus-low return spread is 0.68%, 0.75%, and 0.69% for $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$, and the associated t -statistics are above 5.20. The row labeled “CAPM Alpha” indicates that exposures to market factor cannot explain the return differences between High- and Low-quintiles. The last row reports the annualized Sharpe ratio for each long-short portfolio. All trading strategies have a high Sharpe ratio during the full sample period. Specifically, the Sharpe ratio of the portfolio sorted by the simple $LASSO$ is 1.22, and increases to 1.47, 1.51, and 1.48 when sorted by $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$.

To illuminate how the long-short portfolios based on the four signals perform over time, we compute the cumulative profits for the resulting equal-weighted strategies based on an initial investment of $W_0 = \$1$. Specifically, we compound the daily returns of the spread portfolio sorted by overnight $LASSO$ signals by measuring the cumulative return W_d on day

⁸The total number of sector ETFs per day starts from 24 in January 2000 and rises to 58 in January 2006 (i.e., the beginning of the out-of-sample testing period). The universe further increases to 220 sector ETFs by the end of the sample period in July 2020.

d as follows:

$$W_d = W_{d-1} \times (1 + R_{high,d} - R_{low,d} + R_{rf,d}), \quad d = 1, 2, \dots, \quad (15)$$

where $R_{high,d}$, $R_{low,d}$, and $R_{rf,d}$ are returns on Quintile 5, Quintile 1, and the risk-free rate on day d , respectively.

Fig. 5 plots the trajectory of W_d starting from a given $W_0 = \$1$ initial investment at the start of January 2006. The portfolio value based on all four LASSO strategies continues to rise throughout the entire sample without experiencing major drawdowns. Consistent with Table 8, the two-stage signals generate higher profitability than the simple *LASSO* signal, further highlighting the benefits of adding economic constraints when selecting features.⁹

4.4. *LASSO Signal Combination*

The previous sections show that all four LASSO signals predict sector ETF returns. When there are multiple signals, the next question is then how to combine them for generating a composite signal that improves over any stand-alone signals. We investigate two approaches for signal combination below.

4.4.1. *Simple Averaged Composite Signal*

One common approach for signal combination is to simply take an “average” of them. The main motivation is that different signals protect each other from their individual errors (as long as they do not have all errors in the same direction). Since all four signals have different signal values, we first adjust their signal value to make them comparable. To do so, we cross-sectionally rank each signal period-by-period and map these ranks to the $[0, 1]$ interval. As a result, we have a percentage rank score for each signal. We then take an average

⁹In the Internet Appendix, we perform Monte Carlo simulations to show that machine learning algorithms based on priori knowledge about the firms’ factor loadings have superior performance compared to purely data-driven machine learning algorithms.

of those scores to obtain a new combined signal, namely $Composite^{AVG}$. By construction, the $Composite^{AVG}$ signal ignores the differences in signal strength.

4.4.2. Boosted Composite Signal

Our second signal combination approach uses the Gradient Boosted Regression Trees (GBRT), which is a powerful supervised learning method that has been widely used because it often achieves state-of-the-art prediction performance. We consider the following setup for the signal combination problem. Given an input of multiple signals, e.g., $\mathbf{f}_{j,d,i}^{etf} = (f_{j,d,i}^{etf,LASSO}, f_{j,d,i}^{etf,LASSO_{ADP}}, f_{j,d,i}^{etf,LASSO_{CS}}, f_{j,d,i}^{etf,LASSO_{LF}})$ and the forward returns, e.g., $r_{j,d,i+1}^{etf}$, GBRT uses K trees to predict the output:

$$GBRT_{j,d,i} = \sum_{k=1}^K \mathcal{T}_k(f_{j,d,i}^{etf}), \quad (16)$$

where each $\mathcal{T}_k$ is a single regression tree, which partitions the space of input variables into rectangles and then fits a constant to each rectangle. GBRT then adds multiple regression trees (tree ensemble) to provide a more flexible learning method. By using the additive tree structures, GBRT can automatically capture the nonlinearity and interactions that are challenging for linear models. However, such flexibility could lead to overfitting and thus poor out-of-sample performance. Regularization becomes extremely important in fitting GBRT.

Our implementation of GBRT relies on two mechanisms to reduce overfitting. First, we use the validation set approach to tune two parameters: maximum depth that controls how deep each tree can grow and the learning rate that controls the relative contribution of each tree to the ensemble. The selected maximum depth is typically less than two, indicating no higher-order interactions among predictors. Second, we use early stopping to avoid overfitting the training data. Similar to the LASSO fit in Section 2.2.4, we perform a rolling fit, where each fit uses 1-year, 3-month, and 1-month data as our training, validation and testing samples, respectively. The first training and validation sample periods are from January 2005 to December 2005 and from January 2006 to March 2006, respectively. We call this

composite signal $Composite^{GBRT}$.

4.4.3. Predictive Power of the Composite Signals

We test the two composite signals $Composite^{AVG}$ and $Composite^{GBRT}$ by forming portfolios. Similar to the previous analyses, we focus on overnight composite signals generated at the market open. At 10:00 market open of each trading day d , we sort ETFs into quintile portfolios based on their composite signals. We compute equal-weighted returns on each quintile portfolio and a long-short portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom quintile with low signal values. We hold the portfolios until market close of the same day.

Table 8 summarizes the portfolio sorting results. It shows that $Composite^{AVG}$ has a realized monthly return of 0.73%, and an annualized Sharpe ratio of 1.69 which is a significant improvement over any of the individual LASSO signals. The last column shows that $Composite^{GBRT}$ further improves the performance. The equal-weighted average monthly return increases from -0.20% on Quintile 1 to 0.98% on Quintile 5, yielding a return spread of 1.18% per month with a t -statistic of 7.84 and an annualized Sharpe ratio of 2.34. These numbers suggest that $Composite^{GBRT}$ dominates all other signals economically and statistically, illustrating the advantage of boosted regression tree in combining signals.

Fig. 6 plots the cumulative gain of the long-short portfolios based on different signals including four stand-alone signals and two composite signals. The y -axis presents the dollar value given $W_0 = \$1$ initial investment at the start of April 2006. The portfolio value based on all six strategies continues to rise throughout the entire sample. The $Composite^{GBRT}$ signal continues to generate superior performance during the full sample period. This result is consistent with the previous finding in Table 8 that the $Composite^{GBRT}$ signal has the highest return and Sharpe ratio.

5. Low-Frequency Return Predictability

The ETF predictability shown in previous sections relies on high-frequency LASSO signals trained using high-frequency ETF and stock returns. The fine granularity of our analyses has two main advantages over traditional analyses based on low-frequency data for slower signals. First, our intraday analyses enable us to capture short-lived intraday sector signals which might be missed in low-frequency sector analyses. Second, with larger amount of intraday data, our analyses have higher statistical power that alleviates concerns of false discoveries. However, it is natural to ask whether the structural information remains useful in a traditional low-frequency setting. In fact, we expect structural information to help in the low-frequency setting because introducing lower-dimensional structures usually reduces estimation errors when data are fewer and signal-to-noise ratio is lower. To shed light on this, we measure weekly returns from Tuesday-close to Tuesday-close, and retrain all four LASSO signals using lower-frequency daily and weekly data.¹⁰

Specifically, we follow the same procedures described in Section 2.2 to train four LASSO algorithms for predicting daily (weekly) ETF returns using lagged daily (weekly) stock returns.¹¹ After obtaining the four stand-alone LASSO signals, we use them as inputs to construct two composite signals as described in Section 4.4. As a result, we have two sets of machine-learning (*ML*) signals at daily and weekly frequency, each consisting of six signals, namely $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, $LASSO_{LF}$, $Composite^{AVG}$, and $Composite^{GBRT}$.

To evaluate the predictive performance of the six signals at daily or weekly frequency, we run the following Fama and MacBeth (1973) cross-sectional regressions for each *ML* signal at each frequency t :

$$r_{j,t}^{etf} = \gamma_{0,t} + \gamma_{ML,t} f_{j,t-1}^{etf,ML} + \gamma_{news,t} f_{j,t-1}^{etf,news} + \sum_{k=1}^K \gamma_{k,t} Z_{j,t-1,k} + \epsilon_{j,t}, \quad (17)$$

¹⁰We do not retrain signals using monthly data due to the limitation of effective sample size.

¹¹The time-series average of the cross-sectional correlations of the LASSO signals ranges from 0.14 to 0.39 for daily signals, and from 0.22 to 0.53 for weekly signals.

where $f_{j,t-1}^{etf,news}$ is the lagged daily or weekly news return aggregated from overnight and intraday 30-minute news returns. The control variables $Z_{j,t-1,k}$ for ETF j include lagged daily or weekly ETF return, the Amihud illiquidity measure, realized volatility, ETF price momentum, and ETF size, all of which are measured at the end of day or week $t-1$.

Panels A and B in Table 9 report the estimated coefficients and t -statistics predicting daily and weekly ETF returns using the corresponding LASSO signals. Overall, our key result that $LASSO_{CS}$ and $LASSO_{LF}$ dominate the simple LASSO and adaptive LASSO remains robust for lower-frequency signals. For example, daily $LASSO_{CS}$ and $LASSO_{LF}$ signals from Panel A are associated with higher t -statistics of 3.33 and 3.93 and higher adjusted R^2 's of 23.22% and 23.16%, compared to the t -statistics of 2.69 and 2.87 and the adjusted R^2 's of 21.3% and 22.39% for daily $LASSO$ and $LASSO_{ADP}$ signals. The patterns are similar for weekly LASSO signals in Panel B. For the two composite signals, the simple average signal $Composite^{AVG}$ performs relatively well but does not always beat all stand-alone signals. On the other hand, the composite signal based on boosted regression tree always performs the best, delivering the highest t -statistic and adjusted R^2 at both frequencies. To evaluate the economic significance, we report the rise in annual ETF return associated with a two-standard-deviation increase in various daily and weekly LASSO signals in the table below. At both frequencies, the rise in ETF returns based on $LASSO_{CS}$ or $LASSO_{LF}$ is higher than that based on $LASSO$ or $LASSO_{ADP}$, and $Composite^{GBRT}$ dominates all individual LASSO signals in producing the greatest rise in annual ETF returns.

	<i>Rise in Annual ETF Returns</i>					
	<i>LASSO</i>	<i>LASSO_{ADP}</i>	<i>LASSO_{CS}</i>	<i>LASSO_{LF}</i>	<i>Composite^{AVG}</i>	<i>Composite^{GBRT}</i>
Daily Signal	8.94%	9.59%	13.11%	14.18%	13.32%	15.01%
Weekly Signal	4.16%	4.65%	6.42%	7.31%	6.73%	7.79%

We next form portfolios using low-frequency LASSO signals. At the end of each day (week), we sort ETFs into quintile portfolios based on one of the daily (weekly) LASSO signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom

quintile with low signal values, with a holding period of one day (week). Table 10 reports the monthly portfolio returns in percentage and the annualized Sharpe ratio. Consistent with the regression results, $LASSO_{CS}$ and $LASSO_{LF}$ outperform $LASSO$ and $LASSO_{ADP}$ with higher monthly return spread and higher Sharpe ratio; the composite signal based on GBRT dominates the rest, delivering a monthly return spread of 1.3% and an annual Sharpe ratio of 1.74 at daily frequency, and a monthly spread of 0.85% and an annual Sharpe ratio of 0.96 at weekly frequency. We find that weekly LASSO signals are not as powerful as daily signals, which shows the importance of using higher-frequency data to extract stronger signals. Taken together, the results are robust when we combine structural information with machine learning algorithms using lower-frequency data.

6. Robustness Tests

In this section, we perform various robustness checks controlling for intra-industry lead-lag effect, replacing ETF returns by Fama-French 49 industry portfolios, using firm residual returns to train LASSO signals, and exploring the strategy profitability after accounting for trading costs. The Internet Appendix shows additional robustness tests such as replacing price-based ETF returns by NAV-based ETF returns, predicting ETF returns with Elastic Net, and examining performance on different days of the week or during crisis periods.

6.1. Intra-Industry Lead-Lag Effect

Hou (2007) shows that big firms lead small firms within the same industry, and this intra-industry lead-lag effect drives the overall lead-lag effect. One implication of such an effect is that lagged return on big firms in the same industry of the sector ETF predicts forward ETF returns. To understand whether the big-firm effect exists in our sample, and more importantly, whether our LASSO signals remain intact after controlling for the big-firm effect, we construct another predictor *Big* based on lagged returns of big firms. In particular,

for each sector ETF, *Big* is the equal-weighted average return of top size-decile firms within the same industry of the sector ETF. Following Hou (2007) who uses weekly returns to examine the lead-lag effect, we focus on the comparison of low-frequency LASSO signals and *Big* returns estimated using daily or weekly returns.

Table A.1 summarizes the Fama-MacBeth regression results predicting daily (weekly) ETF returns using various LASSO signals controlling for lagged daily (weekly) *Big* return.¹² Overall, the regression coefficients and *t*-statistics of the different ML signals remain nearly identical to those in Table 9 at both frequencies, further illuminating the robust predictive power of our LASSO signals. In contrast, *Big* is only marginally significant at 10% level in most cases.

6.2. Fama-French 49 Industry Portfolios

Our analyses so far focus on predicting returns of sector ETFs, which are directly tradable at various frequencies. However, most sector ETFs were launched after 2000 and thus have relatively short time series. To examine whether our results remain robust over longer sample, we now switch to Fama-French 49 industry portfolios as our testing assets. We obtain daily returns on 49 equal-weighted industry portfolios from Ken French's Data Library and daily returns on individual stocks from CRSP. To construct machine-learning (*ML*) signals, we follow the same procedures described in Section 2.2 to train four LASSO algorithms for predicting daily industry portfolio returns using lagged daily stock returns. After obtaining the four stand-alone LASSO signals, we use them as inputs to construct two composite signals as described in Section 4.4. Given that analyst coverage data for constructing *LASSO_{LF}* (and the two composite signals) only became available in January 1976, we use the five years of data between 1976 and 1980 as our first training and validation sets, and begin our

¹²Since *Big* is highly correlated with lagged ETF return, we remove the latter from the regressions to avoid multicollinearity. Including lagged residual ETF return orthogonal to *Big* produces quantitatively similar results.

out-of-sample evaluation from January 1981 until July 2020.¹³

To evaluate the predictive performance of the six *ML* signals, we run the following Fama and MacBeth (1973) cross-sectional regressions for each *ML* signal:

$$r_{j,d}^{IP} = \gamma_{0,d} + \gamma_{ML,d} f_{j,d-1}^{IP,ML} + \sum_{k=1}^K \gamma_{k,d} Z_{j,d-1,k} + \epsilon_{j,d}, \quad (18)$$

where $r_{j,d}^{IP}$ is the equal-weighted return of industry portfolio j on day d ; the control variables $Z_{j,d-1,k}$ for industry j include lagged daily return (*RET*), the Amihud illiquidity measure (*ILLIQ*), realized volatility (*RVOL*), price momentum (*Mom*), and industry size (*Size*), all of which are measured at the end of day $d - 1$.¹⁴ *RET*, *RVOL*, and *Mom* are calculated based on daily industry portfolio returns, while industry *ILLQ* and *Size* are constructed by equally averaging the corresponding measure of stocks in a given industry.

Panel A of Table A.2 reports the regression results from predicting daily industry portfolio returns using *ML* signals. Overall, the two LASSO signals based on structural information yield stronger predictive power than LASSO and adaptive LASSO. For instance, the t -statistics for $LASSO_{CS}$ and $LASSO_{LF}$ coefficients are 5.76 and 6.01 compared to the t -statistics of 4.74 and 5.05 for *LASSO* and $LASSO_{ADP}$, and the adjusted R^2 's for $LASSO_{CS}$ and $LASSO_{LF}$ are 14.29% and 14.47%, higher than the R^2 's of 14.01% and 14.02% generated by *LASSO* and $LASSO_{ADP}$. The simple average composite signal performs on par with $LASSO_{LF}$, and the composite signal based on boosted regression tree has the highest t -statistic of 7.12 and adjusted R^2 of 15.21%.

We next form portfolios using *ML* signals. At the end of each day, we sort 49 industry portfolios into quintile portfolios based on daily *ML* signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys industry portfolios in

¹³We rely on eight BEA IO Surveys (1972, 1977, 1982, 1987, 1992, 1997, 2002, and 2007) on a rolling basis for constructing the $LASSO_{CS}$ signal. Specifically, to train LASSO signals, we use the most recent IO Use Table prior to the beginning of each 6-year implementation period (i.e., 4-year training, 1-year validation, and 1-year testing). For example, we use the 1972 IO Survey for our first implementation period from January 1976 to December 1981.

¹⁴We no longer include the *News* predictor because the RavenPack news data are not available prior to 2000.

the top quintile with high signal values and sells industry portfolios in the bottom quintile with low signal values, with a holding period of one day. Panel B in Table A.2 reports the monthly portfolio returns in percentage and the annualized Sharpe ratio. Consistent with the regression results, the two economic-link-based LASSO signals $LASSO_{CS}$ and $LASSO_{LF}$ outperform $LASSO$ and $LASSO_{ADP}$ with higher monthly return spread and higher Sharpe ratio, and the composite signal based on GBRT outperforms the rest. Taken together, our main conclusion that imposing structural information can improve machine learning performance holds over the extended sample period.

6.3. Using Firm Residual Returns to Train the LASSO Signal

Our LASSO signals are trained from total returns of individual stocks. The implicit assumption is that individual stock price movements mainly capture firm-level granular information. However, if firm returns are exposed to aggregate risk (e.g., market), the documented predictive power of the LASSO signals may be driven by macro-level information. To better understand the source of the predictability, we re-construct the simple LASSO signal using firm residual returns as predictors. Specifically, we obtain firm residual returns as follows:

$$res_{s,d,i}^{stc} = r_{s,d,i}^{stc} - \beta_{s,d-1}^{mkt} r_{d,i}^{mkt}, \quad (19)$$

where $res_{s,d,i}^{stc}$ and $r_{s,d,i}^{stc}$ are the residual and total returns of stock s on day d in interval i , respectively; $r_{d,i}^{mkt}$ is market (SPY) return on day d in interval i ; $\beta_{s,d-1}^{mkt}$ is the market beta of stock s on day $d - 1$, estimated by the coefficient from regressing daily stock returns on daily market returns using one-year rolling window. Then we retrain the simple LASSO signal using residual returns following the procedures detailed in Section 2.2.1 and name it $LASSO_{RES}$.

Table A.3 reports the Fama-MacBeth regression results predicting each 30-minute ETF returns using $LASSO_{RES}$; the last column presents the average coefficients, t -statistics, and adjusted R^2 across different 30-minute intervals. The average coefficient of $LASSO_{RES}$ is

0.170 with a t -statistic of 10.71; the average adjusted- R^2 is 14.94%. The corresponding figures for the simple *LASSO* signal trained from total returns in Table 2 are 0.143, 8.46, and 14.51%, respectively. Thus, the *LASSO* signal trained from residual returns results in similar and even stronger predictability, further corroborating the idea that the predictive power of our *LASSO* signals mainly arises from firm-level granular information.

6.4. Trading Costs

In this subsection, we evaluate the profitability of tradings strategies based on the information signals from machine learning algorithms after considering the impact of transaction costs on trading profits. To mitigate the erosion of trading costs on performance, we follow the idea in Chinco, Clark-Joseph, and Ye (2019), trading sector ETFs only when the implied profit based on the information signal exceeds the bid-ask spread. Specifically, we compute the proportional effective spread for each ETF in our sample as the scaled difference between price and the midpoint of NBBO (National Best Bid and Offer), and use the sample average of 16 bps as the bid-ask spread for all ETFs. Conditional on a trade being executed, we can calculate the strategy's realized returns net of the bid-ask spread as $r_{j,d}^{etf} \cdot \text{Sign}[f_{j,d}^{etf,L}] - \text{spread}$, where L stands for a given machine learning predictor. Finally, to form the portfolio, we define the fraction of investment on the j th ETF as follows:

$$w_{j,d}^{etf} = \frac{\left(1/\sigma_{j,d}^{etf,L}\right) \cdot f_{j,d}^{etf,L} \cdot 1_{|f_{j,d}^{etf,L}| > \text{spread}}}{\sum_j \left| \left(1/\sigma_{j,d}^{etf,L}\right) \cdot f_{j,d}^{etf,L} \cdot 1_{|f_{j,d}^{etf,L}| > \text{spread}} \right|}, \quad (20)$$

where $\sigma_{j,d}^{etf,L}$ stands for the standard deviation of machine learning predictor L , estimated using the previous 21-day forecasts. Thus, the net return to our strategy based on machine-learning predictor L is calculated by

$$R_d^L = \sum_j \left(r_{j,d}^{etf} \cdot \text{Sign}[f_{j,d}^{etf,L}] - \text{spread} \right) \times w_{j,d}^{etf}. \quad (21)$$

Table A.4 shows the transaction cost-adjusted returns along with the annualized Sharpe ratios for each trading strategy based on different machine-learning predictors. Notably, all strategies generate a significant positive excess return, indicating the trading strategy is profitable after transaction costs. Specifically, the strategy based on LASSO yields an average monthly return of 0.31% with an annual Sharpe ratio of 0.59. After incorporating structural information of economic links, the performance increases to 0.42% (0.40%) with a Sharpe ratio of 0.84 (0.82) for $LASSO_{CS}$ ($LASSO_{LF}$). The strategy based on $Composite^{GBRT}$ yields the highest return of 0.64% and the highest Sharpe ratio of 1.30, further emphasizing the advantage of boosted regression tree in combining signals. Overall, the machine learning predictors can generate a significant economic gain even after considering trading costs.

7. Conclusion

We show in this paper a strong link between granular information contained in individual stock prices and sectoral movements. We find that a predictor that aggregates price movements of the cross section of the US equity market can strongly predict future sector ETF returns. When we incorporate structural information of economic links among firms, the resulting information signals have even stronger predictability. These results support the hypothesis of granular origins of aggregate shocks.

On the methodological front, our study illuminates the advantages of combining big data with structural knowledge over purely data-driven techniques. Both our empirical tests and Monte Carlo simulations support this view. We believe that integrating big data with economic theory provides a fruitful avenue for future research.

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Table 1 Descriptive statistics

This table reports the descriptive statistics of our main variables. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. Panel A reports the time-series average of the cross-sectional mean, standard deviation, and quantiles of each variable. Panel B reports the time-series average of the cross-sectional correlations of these variables. All LASSO signals presented in this table are overnight signals corresponding to interval $i = 1$ in Eq. (9). $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. $News$ is the overnight news-driven signal corresponding to interval $i = 1$ in Eq. (10). RET is the overnight ETF return. $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, $LASSO_{LF}$, $News$, and RET are all reported in bps per day. $ILLIQ$ is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. $RVOL$ is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. Mom for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. $Size$ is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily.

Panel A: Cross-Sectional Summary Statistics

	Mean	Std	P1	P25	Median	P75	P99
$LASSO$ (bps)	-0.31	8.96	-22.18	-3.56	-0.16	3.04	21.01
$LASSO_{ADP}$ (bps)	-0.36	6.87	-15.88	-1.86	-0.27	1.17	14.94
$LASSO_{CS}$ (bps)	-0.34	7.16	-19.72	-2.09	-0.20	1.41	17.19
$LASSO_{LF}$ (bps)	-0.36	6.47	-15.50	-1.74	-0.19	0.96	15.68
$News$ (bps)	4.03	165.07	-424.31	-50.66	4.96	61.34	414.37
RET (bps)	1.54	57.86	-127.13	-28.61	1.91	31.90	131.44
$ILLIQ$	-20.37	2.37	-25.51	-21.73	-20.16	-18.73	-16.10
$RVOL$	22.02%	11.60%	9.73%	15.70%	19.37%	25.00%	59.33%
Mom	-0.10%	19.53%	-48.59%	-8.84%	-0.05%	8.68%	47.89%
$Size$	6.18	1.43	3.42	5.20	6.20	7.08	9.09

Panel B: Cross-Sectional Correlations

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$News$	RET	$ILLIQ$	$RVOL$	Mom	$Size$
$LASSO$	1.000	0.461	0.283	0.241	0.013	0.028	0.002	-0.031	0.001	0.001
$LASSO_{ADP}$	0.461	1.000	0.380	0.365	0.005	0.015	0.010	-0.052	0.004	-0.007
$LASSO_{CS}$	0.283	0.380	1.000	0.538	0.028	0.058	0.021	-0.052	0.001	-0.016
$LASSO_{LF}$	0.241	0.365	0.538	1.000	0.030	0.070	0.019	-0.057	0.002	-0.011
$News$	0.013	0.005	0.028	0.030	1.000	0.440	-0.002	-0.003	0.001	0.005
RET	0.028	0.015	0.058	0.070	0.440	1.000	-0.004	0.005	-0.001	0.007
$ILLIQ$	0.002	0.010	0.021	0.019	-0.002	-0.004	1.000	0.183	0.000	-0.896
$RVOL$	-0.031	-0.052	-0.052	-0.057	-0.003	0.005	0.183	1.000	-0.001	-0.226
Mom	0.001	0.004	0.001	0.002	0.001	-0.001	0.000	-0.001	1.000	0.000
$Size$	0.001	-0.007	-0.016	-0.011	0.005	0.007	-0.896	-0.226	0.000	1.000

Table 2 Predicting 30-minute ahead ETF returns: high-frequency LASSO signal

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting every intraday half-hour ETF return in interval i ($i = 2, \dots, 13$) on day d by the *LASSO* and *News* signals calculated using inputs from interval $i - 1$. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. *LASSO* is the simple LASSO signal from Eq. (9). *News* is the news-driven signal from Eq. (10). *RET* is the lagged overnight or 30-minute ETF return. *ILLQ* is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. *RVOL* is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. *Mom* for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. *Size* is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return Interval													
Start	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	Mean
End	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	
Intercept	-0.071 (-0.06)	2.199 (2.14)	1.318 (1.49)	0.006 (0.01)	0.697 (0.93)	0.360 (0.51)	-0.338 (-0.48)	0.018 (0.03)	-1.056 (-1.31)	-1.882 (-1.13)	1.730 (2.15)	-1.064 (-1.16)	0.160 (0.26)
<i>LASSO</i>	0.101 (6.28)	0.170 (8.94)	0.188 (10.86)	0.136 (8.51)	0.169 (10.29)	0.178 (11.50)	0.160 (8.98)	0.139 (7.65)	0.124 (7.16)	0.129 (9.11)	0.117 (5.95)	0.107 (6.29)	0.143 (8.46)
<i>News</i>	0.021 (21.06)	0.089 (1.68)	-0.066 (-0.98)	0.144 (1.17)	0.018 (0.69)	0.002 (0.17)	0.059 (1.11)	-0.074 (-1.42)	0.105 (0.91)	0.033 (3.07)	0.008 (0.63)	0.582 (0.87)	0.077 (2.41)
RET	-0.088 (-26.39)	-0.092 (-20.56)	-0.112 (-25.26)	-0.113 (-23.81)	-0.131 (-28.84)	-0.141 (-32.73)	-0.147 (-29.93)	-0.159 (-31.84)	-0.173 (-32.11)	-0.169 (-35.57)	-0.175 (-36.18)	-0.215 (-42.01)	-0.143 (-30.43)
ILLIQ	0.023 (0.26)	0.114 (1.67)	0.001 (0.01)	-0.050 (-0.95)	0.019 (0.37)	0.039 (0.81)	-0.024 (-0.47)	0.013 (0.28)	-0.016 (-0.32)	-0.140 (-1.17)	0.044 (0.71)	-0.039 (-0.64)	-0.001 (0.05)
RVOL	-2.604 (-1.65)	-0.723 (-0.46)	-2.221 (-1.68)	-0.907 (-0.82)	-0.202 (-0.19)	-0.147 (-0.15)	-2.214 (-2.18)	-2.521 (-2.63)	-0.377 (-0.34)	0.797 (0.79)	-0.594 (-0.61)	1.719 (1.67)	-0.833 (-0.69)
Mom	0.747 (1.52)	0.152 (0.36)	0.176 (0.51)	-0.083 (-0.27)	-0.321 (-1.13)	0.501 (1.66)	-0.683 (-2.52)	-0.110 (-0.41)	0.313 (1.08)	0.524 (2.00)	-0.188 (-0.69)	-0.179 (-0.57)	0.071 (0.13)
Size	0.086 (0.57)	-0.029 (-0.24)	-0.131 (-1.27)	-0.157 (-1.55)	-0.059 (-0.67)	0.063 (0.75)	-0.007 (-0.08)	0.012 (0.14)	0.056 (0.65)	-0.129 (-0.83)	-0.073 (-0.71)	-0.019 (-0.16)	-0.032 (-0.28)
Adj- R^2 (%)	15.96	14.79	14.95	14.69	13.89	13.53	13.67	14.05	14.85	15.22	15.17	13.37	14.51

Table 3 Predicting ETF returns from 10:00 to different ending time of the same day: overnight LASSO signal

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting short-term ETF returns from 10:00 to different ending time of the same day by the overnight *LASSO* and *News* signals. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. *LASSO* is the overnight LASSO signal corresponding to interval $i = 1$ in Eq. (9). *News* is the overnight news-driven signal corresponding to interval $i = 1$ in Eq. (10). *RET* is the overnight ETF return. *ILLQ* is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. *RVOL* is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. *Mom* for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. *Size* is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return Interval													
Start	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	Mean
End	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	
Intercept	-0.071 (-0.06)	2.261 (1.76)	2.882 (2.08)	2.436 (1.64)	3.472 (2.22)	3.581 (2.20)	3.414 (2.06)	4.109 (2.33)	3.164 (1.78)	2.213 (0.95)	5.037 (2.67)	4.409 (2.30)	3.076 (1.83)
<i>LASSO</i>	0.101 (6.28)	0.104 (5.25)	0.104 (4.65)	0.116 (4.75)	0.114 (4.49)	0.108 (4.13)	0.119 (4.34)	0.112 (3.95)	0.125 (4.31)	0.148 (4.68)	0.141 (4.44)	0.121 (3.79)	0.118 (4.59)
<i>News</i>	0.021 (21.06)	0.023 (19.23)	0.023 (17.23)	0.024 (16.87)	0.025 (16.85)	0.025 (16.44)	0.025 (15.55)	0.025 (15.66)	0.026 (15.35)	0.027 (14.85)	0.027 (15.16)	0.028 (14.93)	0.025 (16.60)
RET	-0.088 (-26.39)	-0.098 (-22.13)	-0.100 (-19.92)	-0.098 (-18.33)	-0.099 (-17.49)	-0.100 (-17.24)	-0.098 (-16.06)	-0.101 (-16.04)	-0.098 (-15.72)	-0.100 (-15.27)	-0.098 (-14.98)	-0.104 (-15.66)	-0.099 (-17.94)
ILLIQ	0.023 (0.26)	0.166 (1.72)	0.158 (1.52)	0.076 (0.66)	0.133 (1.08)	0.151 (1.17)	0.089 (0.67)	0.111 (0.82)	0.086 (0.62)	-0.039 (-0.22)	0.126 (0.84)	0.083 (0.56)	0.097 (0.81)
RVOL	-2.604 (-1.65)	-3.548 (-1.59)	-5.685 (-2.25)	-6.592 (-2.41)	-7.636 (-2.60)	-6.987 (-2.30)	-8.966 (-2.85)	-10.920 (-3.34)	-11.349 (-3.44)	-10.779 (-3.11)	-10.748 (-3.03)	-9.272 (-2.54)	-7.924 (-2.59)
Mom	0.747 (1.52)	0.733 (1.20)	1.017 (1.56)	1.093 (1.51)	0.674 (0.89)	1.234 (1.55)	0.670 (0.82)	0.632 (0.75)	1.077 (1.22)	1.412 (1.54)	1.203 (1.33)	1.008 (1.08)	0.958 (1.25)
Size	0.086 (0.57)	0.106 (0.60)	0.038 (0.19)	-0.124 (-0.55)	-0.099 (-0.42)	-0.036 (-0.14)	-0.167 (-0.64)	-0.204 (-0.76)	-0.173 (-0.62)	-0.358 (-1.12)	-0.203 (-0.67)	-0.281 (-0.91)	-0.118 (-0.37)
Adj- R^2 (%)	15.96	17.62	18.07	17.93	18.06	18.15	18.31	18.42	18.45	18.50	18.53	18.36	18.03

Table 4 Predicting ETF returns from 10:00 on day d to 16:00 on day $d+k$: overnight LASSO signal

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting cumulative ETF returns from 10:00 on day d to 16:00 on day $d+k$ ($k = 0, 1, \dots, 4$) by the overnight *LASSO* and *News* signals. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. *LASSO* is the overnight LASSO signal corresponding to interval $i = 1$ in Eq. (9). *News* is the overnight news-driven signal corresponding to interval $i = 1$ in Eq. (10). *RET* is the overnight ETF return. *ILLQ* is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. *RVOL* is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. *Mom* for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. *Size* is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Start End	Return Interval					Mean
	10:00 on day d 16:00 on day d	10:00 on day d 16:00 on day $d+1$	10:00 on day d 16:00 on day $d+2$	10:00 on day d 16:00 on day $d+3$	10:00 on day d 16:00 on day $d+4$	
Intercept	4.409 (2.30)	11.792 (3.02)	18.558 (3.16)	24.235 (3.20)	32.278 (3.51)	18.255 (3.04)
<i>LASSO</i>	0.121 (3.79)	0.135 (2.26)	0.147 (1.88)	0.147 (1.85)	0.203 (1.82)	0.151 (2.32)
<i>News</i>	0.028 (14.93)	0.029 (9.15)	0.032 (7.98)	0.032 (6.77)	0.032 (5.84)	0.031 (8.93)
RET	-0.104 (-15.66)	-0.073 (-6.10)	-0.084 (-5.51)	-0.088 (-4.84)	-0.079 (-3.64)	-0.086 (-7.15)
ILLIQ	0.083 (0.56)	0.176 (0.58)	0.516 (1.09)	0.758 (1.23)	1.211 (1.61)	0.549 (1.01)
RVOL	-9.272 (-2.54)	-21.384 (-2.97)	-27.002 (-2.62)	-31.948 (-2.36)	-36.290 (-2.23)	-25.179 (-2.54)
Mom	1.008 (1.08)	-0.129 (-0.08)	-0.599 (-0.28)	-0.854 (-0.34)	-1.477 (-0.50)	-0.410 (-0.02)
Size	-0.281 (-0.91)	-0.346 (-0.52)	0.275 (0.26)	0.847 (0.62)	1.698 (1.01)	0.439 (0.09)
Adj- R^2 (%)	18.36	18.40	17.93	17.63	17.47	17.96

Table 5 Predicting 30-minute ahead ETF returns: alternative high-frequency LASSO signals

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting every intraday half-hour ETF return in interval i ($i = 2, \dots, 13$) on day d by the alternative *LASSO* and the *News* signals, all calculated using inputs from interval $i - 1$. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All alternative LASSO signals are defined by Eq. (9). *LASSO_{ADP}* is the adaptive LASSO signal, *LASSO_{CS}* is the LASSO signal based on customer-supplier relation, and *LASSO_{LF}* is the LASSO signal based on leading firms. *News* is the news-driven signal from Eq. (10). All regressions control for *RET*, *ILLQ*, *RVOL*, *Mom*, and *Size*, all of which are defined in Table 2. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return Interval													
Start	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	Mean
End	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	
Panel A: Adaptive LASSO													
Intercept	-0.366 (-0.32)	1.470 (1.44)	1.365 (1.56)	-0.085 (-0.11)	0.624 (0.83)	0.351 (0.50)	-0.466 (-0.65)	-0.145 (-0.21)	-0.880 (-1.09)	-1.863 (-1.11)	1.430 (1.79)	-0.856 (-0.96)	0.048 (0.14)
<i>LASSO_{ADP}</i>	0.175 (6.08)	0.234 (9.77)	0.244 (10.66)	0.178 (8.57)	0.222 (9.55)	0.203 (9.34)	0.174 (7.71)	0.181 (8.46)	0.158 (6.93)	0.147 (7.98)	0.104 (5.48)	0.139 (5.96)	0.180 (8.04)
<i>News</i>	0.021 (20.73)	0.089 (1.70)	-0.048 (-0.86)	0.137 (1.21)	0.016 (0.61)	0.006 (0.58)	0.048 (1.27)	-0.073 (-1.41)	0.090 (0.69)	0.034 (3.13)	0.010 (0.81)	0.529 (0.86)	0.072 (2.44)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	17.62	16.36	16.57	16.23	15.68	15.26	15.39	15.83	16.72	16.98	16.93	15.21	16.23
Panel B: Customer-Supplier-Based LASSO													
Intercept	-0.336 (-0.30)	1.348 (1.35)	0.997 (1.17)	-0.031 (-0.04)	0.407 (0.54)	0.325 (0.46)	-0.665 (-0.94)	-0.151 (-0.23)	-0.997 (-1.23)	-1.792 (-1.11)	1.177 (1.51)	-0.628 (-0.67)	-0.029 (0.04)
<i>LASSO_{CS}</i>	0.212 (7.08)	0.284 (11.02)	0.237 (10.45)	0.302 (13.81)	0.244 (10.28)	0.279 (13.68)	0.249 (10.97)	0.240 (11.15)	0.242 (10.61)	0.218 (12.05)	0.159 (7.99)	0.169 (7.80)	0.236 (10.57)
<i>News</i>	0.021 (20.86)	0.083 (1.60)	-0.060 (-0.89)	0.117 (1.15)	0.014 (0.59)	0.006 (0.61)	0.059 (1.11)	-0.082 (-1.46)	0.072 (0.55)	0.031 (2.97)	0.010 (0.79)	0.569 (0.96)	0.070 (2.40)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	17.72	16.73	16.98	16.72	15.94	15.50	15.70	16.18	17.02	17.22	17.13	15.53	16.53
Panel C: Leading-Firm-Based LASSO													
Intercept	-0.624 (-0.56)	1.404 (1.44)	1.355 (1.58)	-0.523 (-0.70)	0.525 (0.71)	0.516 (0.73)	-0.242 (-0.34)	-0.107 (-0.16)	-0.729 (-0.90)	-1.702 (-1.09)	1.323 (1.67)	-0.753 (-0.82)	0.037 (0.13)
<i>LASSO_{LF}</i>	0.210 (6.01)	0.364 (13.76)	0.393 (16.36)	0.374 (16.52)	0.349 (15.63)	0.411 (18.51)	0.386 (15.76)	0.329 (15.00)	0.311 (14.48)	0.291 (14.57)	0.226 (10.84)	0.247 (10.69)	0.324 (14.01)
<i>News</i>	0.021 (20.69)	0.076 (1.55)	-0.057 (-0.92)	0.107 (1.29)	0.020 (0.78)	0.007 (0.64)	0.041 (1.15)	-0.077 (-1.43)	0.081 (0.67)	0.034 (3.21)	0.012 (0.93)	0.588 (0.93)	0.071 (2.46)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	18.04	16.93	17.21	16.99	16.15	16.01	16.01	16.32	17.21	17.49	17.35	15.68	16.78

Table 6 Predicting ETF returns from 10:00 to different ending time of the same day: alternative overnight LASSO signals

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting short-term ETF returns from 10:00 to different ending time of the same day by the overnight LASSO and the overnight *News* signals. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All alternative overnight LASSO signals correspond to interval $i = 1$ in Eq. (9). $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. *News* is the overnight news-driven signal corresponding to interval $i = 1$ in Eq. (10). All regressions control for *RET*, *ILLQ*, *RVOL*, *Mom*, and *Size*, all of which are defined in Table 3. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return Interval													
Start	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	10:00	Mean
End	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	
Panel A: Adaptive LASSO													
Intercept	-0.366 (-0.32)	2.154 (1.65)	2.528 (1.80)	2.244 (1.50)	3.179 (2.00)	3.376 (2.04)	3.322 (1.97)	4.024 (2.25)	3.084 (1.72)	2.083 (0.90)	4.910 (2.59)	4.093 (2.14)	2.886 (1.69)
$LASSO_{ADP}$	0.175 (6.08)	0.197 (5.31)	0.230 (5.42)	0.248 (5.29)	0.240 (5.02)	0.235 (4.88)	0.229 (4.57)	0.220 (4.37)	0.229 (4.35)	0.248 (4.45)	0.248 (4.49)	0.211 (3.70)	0.226 (4.83)
<i>News</i>	0.021 (20.73)	0.023 (19.22)	0.023 (17.29)	0.024 (16.63)	0.025 (16.66)	0.025 (16.33)	0.024 (15.32)	0.025 (15.27)	0.026 (14.96)	0.026 (14.46)	0.027 (14.81)	0.027 (14.48)	0.025 (16.35)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	17.62	19.23	19.56	19.39	19.45	19.54	19.75	19.88	19.92	19.96	19.97	19.83	19.51
Panel B: Customer-Supplier-Based LASSO													
Intercept	-0.336 (-0.30)	1.809 (1.39)	2.177 (1.55)	2.085 (1.39)	3.073 (1.94)	3.454 (2.09)	3.388 (2.02)	4.211 (2.37)	3.406 (1.90)	2.305 (0.99)	5.145 (2.73)	4.460 (2.32)	2.931 (1.70)
$LASSO_{CS}$	0.212 (7.08)	0.230 (6.51)	0.287 (6.99)	0.316 (7.10)	0.322 (7.14)	0.302 (6.53)	0.322 (6.80)	0.323 (6.61)	0.318 (6.35)	0.292 (5.53)	0.314 (5.98)	0.289 (5.34)	0.294 (6.50)
<i>News</i>	0.021 (20.86)	0.023 (19.01)	0.023 (17.12)	0.024 (16.49)	0.024 (16.51)	0.024 (16.05)	0.024 (15.19)	0.025 (15.15)	0.026 (15.00)	0.026 (14.48)	0.027 (14.70)	0.028 (14.46)	0.025 (16.25)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	17.72	19.51	19.84	19.62	19.67	19.76	19.98	20.08	20.10	20.09	20.08	19.91	19.70
Panel C: Leading-Firm-Based LASSO													
Intercept	-0.624 (-0.56)	1.356 (1.04)	1.589 (1.14)	1.258 (0.85)	2.218 (1.41)	2.518 (1.54)	2.442 (1.46)	3.330 (1.89)	2.557 (1.44)	1.262 (0.55)	3.986 (2.12)	3.446 (1.82)	2.112 (1.22)
$LASSO_{LF}$	0.210 (6.01)	0.312 (7.43)	0.379 (7.60)	0.392 (7.06)	0.392 (6.80)	0.366 (6.18)	0.365 (5.94)	0.368 (5.72)	0.385 (5.69)	0.369 (5.27)	0.373 (5.23)	0.340 (4.70)	0.354 (6.14)
<i>News</i>	0.021 (20.69)	0.023 (18.84)	0.023 (16.92)	0.024 (16.43)	0.024 (16.33)	0.024 (15.94)	0.024 (15.02)	0.025 (15.01)	0.026 (14.74)	0.026 (14.28)	0.027 (14.66)	0.027 (14.33)	0.025 (16.10)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	18.04	19.65	20.00	19.85	19.89	19.93	20.11	20.27	20.27	20.29	20.29	20.13	19.89

Table 7 Predicting ETF returns from 10:00 on day d to 16:00 on day $d+k$: alternative overnight LASSO signals

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting cumulative ETF returns from 10:00 on day d to 16:00 on day $d+k$ ($k = 0, 1, \dots, 4$) by the alternative overnight *LASSO* and the overnight *News* signals. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All alternative overnight LASSO signals correspond to interval $i = 1$ in Eq. (9). *LASSO_{ADP}* is the adaptive LASSO signal, *LASSO_{CS}* is the LASSO signal based on customer-supplier relation, and *LASSO_{LF}* is the LASSO signal based on leading firms. *News* is the overnight news-driven signal corresponding to interval $i = 1$ in Eq. (10). All regressions control for *RET*, *ILLQ*, *RVOL*, *Mom*, and *Size*, all of which are defined in Table 4. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return interval						
Start	10:00 on day d	10:00 on day d	10:00 on day d	10:00 on day d	10:00 on day d	Mean
End	16:00 on day d	16:00 on day $d+1$	16:00 on day $d+2$	16:00 on day $d+3$	16:00 on day $d+4$	
Panel A: Adaptive LASSO						
Intercept	4.093 (2.14)	10.845 (2.73)	16.828 (2.82)	23.149 (2.99)	30.732 (3.27)	17.129 (2.79)
$LASSO_{ADP}$	0.211 (3.90)	0.236 (2.66)	0.356 (2.44)	0.288 (2.32)	0.335 (2.55)	0.285 (2.77)
$News$	0.027 (14.48)	0.029 (8.94)	0.033 (8.03)	0.032 (6.58)	0.031 (5.56)	0.030 (8.72)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	19.83	19.92	19.39	18.93	18.76	19.37
Panel B: Customer-Supplier-Based LASSO						
Intercept	4.460 (2.32)	10.949 (2.76)	17.292 (2.90)	23.803 (3.06)	31.205 (3.29)	17.542 (2.86)
$LASSO_{CS}$	0.289 (5.34)	0.294 (3.74)	0.190 (2.21)	0.190 (1.99)	0.229 (2.18)	0.238 (3.09)
$News$	0.028 (14.46)	0.028 (8.79)	0.033 (8.00)	0.032 (6.39)	0.031 (5.43)	0.030 (8.61)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	19.91	19.93	19.53	19.06	18.86	19.46
Panel C: Leading-Firm-Based LASSO						
Intercept	3.446 (1.82)	10.147 (2.57)	16.265 (2.70)	22.077 (2.83)	28.754 (3.03)	16.138 (2.59)
$LASSO_{LF}$	0.340 (5.70)	0.426 (3.62)	0.430 (2.57)	0.465 (2.38)	0.450 (2.23)	0.422 (3.30)
$News$	0.027 (14.33)	0.029 (9.09)	0.033 (7.99)	0.031 (6.31)	0.030 (5.24)	0.030 (8.59)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2 (%)	20.13	20.20	19.86	19.46	19.24	19.78

Table 8 Performance of quintile portfolios sorted on the basis of different overnight LASSO signals

This table reports the performance of quintile portfolios formed on the basis of different overnight LASSO signals over the entire out-of-sample period between January 2006 and July 2020. At 10:00 of each trading day d , we sort ETFs into quintile portfolios based on their overnight $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom quintile with low signal values, with a holding period from 10:00 on day d until market close of the same day. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All overnight LASSO signals correspond to interval $i = 1$ in Eq. (9). $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). We multiply daily returns by 21 to get monthly returns in percentage. The row labeled “High-Low” reports the average monthly return spread between Quintiles 5 and 1, and the row labeled “CAPM Alpha” reports the monthly return spread adjusted by market exposure. The row labeled “Annual SR” reports the annualized Sharpe ratio for each long-short portfolio.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
1 (Low)	0.04	-0.01	0.02	-0.01	0.01	-0.20
2	0.32	0.45	0.30	0.40	0.30	0.30
3	0.42	0.43	0.33	0.46	0.49	0.43
4	0.57	0.47	0.55	0.47	0.55	0.49
5 (High)	0.65	0.67	0.77	0.68	0.74	0.98
High-Low	0.61	0.68	0.75	0.69	0.73	1.18
	(4.52)	(5.32)	(5.47)	(5.29)	(6.25)	(7.84)
CAPM Alpha	0.62	0.69	0.76	0.70	0.74	1.19
	(4.60)	(5.46)	(5.57)	(5.44)	(6.33)	(8.00)
Annual SR	1.22	1.47	1.51	1.48	1.69	2.34

Table 9 Predicting daily or weekly ETF returns: low-frequency LASSO signals

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting low-frequency daily or weekly ETF returns by the corresponding machine-learning (ML) signals learned from lagged daily or weekly stock returns. Both daily and weekly (Tuesday-to-Tuesday) returns are calculated using close prices. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. The construction of the low-frequency ML signals is similar to that of the high-frequency signals, except that we use daily (weekly) stock returns to train different LASSO algorithms for predicting daily (weekly) ETF returns. ML signals include $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). $News$ is the lagged daily (Panel A) or weekly (Panel B) news return aggregated from overnight and intraday 30-minute news returns. RET is the lagged daily (Panel A) or weekly (Panel B) ETF return. $ILLQ$ is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. $RVOL$ is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. Mom for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. $Size$ is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: Predicting Daily ETF Returns</i>						
Intercept	6.834 (2.38)	6.188 (2.20)	6.038 (2.16)	5.521 (2.05)	5.608 (2.14)	5.450 (2.04)
ML	0.078 (2.69)	0.108 (2.87)	0.234 (3.33)	0.254 (3.93)	0.240 (3.94)	0.290 (4.23)
$News$	0.004 (2.26)	0.005 (2.55)	0.004 (2.00)	0.005 (2.34)	0.005 (2.24)	0.004 (2.32)
RET	-0.013 (-1.99)	-0.011 (-1.63)	-0.015 (-2.09)	-0.015 (-2.12)	-0.012 (-2.19)	-0.011 (-2.08)
$ILLIQ$	0.221 (0.98)	0.313 (1.26)	0.253 (1.13)	0.278 (1.23)	0.216 (1.21)	0.210 (1.03)
$RVOL$	-10.072 (-1.83)	-9.001 (-1.67)	-8.322 (-1.45)	-11.379 (-1.98)	-11.306 (-1.95)	-10.306 (-1.95)
Mom	2.573 (1.40)	2.253 (1.17)	2.556 (1.33)	1.915 (1.02)	1.924 (1.04)	1.902 (1.01)
$Size$	0.223 (0.46)	0.481 (0.94)	0.182 (0.37)	0.262 (0.55)	0.265 (0.58)	0.252 (0.52)
Adj- R^2 (%)	21.30	22.39	23.22	23.16	23.16	23.58
<i>Panel B: Predicting Weekly ETF Returns</i>						
Intercept	45.874 (3.17)	51.146 (3.66)	50.514 (3.57)	47.892 (3.33)	48.179 (3.42)	47.358 (3.07)
ML	0.080 (1.91)	0.090 (2.17)	0.125 (2.36)	0.143 (2.61)	0.133 (2.73)	0.152 (2.81)
$News$	0.009 (1.15)	0.008 (0.99)	0.008 (0.92)	0.009 (0.99)	0.009 (0.91)	0.009 (1.02)
RET	0.021 (1.55)	0.025 (1.76)	0.018 (1.29)	0.021 (1.54)	0.021 (1.57)	0.022 (1.57)
$ILLIQ$	1.569 (1.60)	2.004 (2.14)	1.856 (2.02)	1.651 (1.83)	1.673 (1.85)	1.657 (1.84)
$RVOL$	-48.067 (-2.18)	-42.272 (-1.90)	-39.913 (-1.80)	-41.429 (-1.85)	-41.866 (-1.90)	-41.458 (-1.87)
Mom	5.463 (0.48)	6.430 (0.82)	7.109 (0.86)	6.105 (0.52)	6.376 (0.59)	6.116 (0.53)
$Size$	2.552 (1.21)	3.397 (1.64)	2.924 (1.44)	2.826 (1.49)	2.301 (1.28)	2.630 (1.46)
Adj- R^2 (%)	21.77	22.34	22.36	22.51	22.32	22.73

Table 10 Performance of quintile portfolios sorted on the basis of different low-frequency LASSO signals

This table reports the performance of quintile portfolios formed on the basis of low-frequency daily or weekly machine-learning (ML) signals over the entire out-of-sample period between January 2006 and July 2020. Panel A reports the results of daily rebalanced portfolio sorts. At the market close of each trading day d , we sort ETFs into quintile portfolios based on daily ML signals which include $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom quintile with low signal values, with a holding period from market close on day d until market close of the next day. Panel B reports the results of weekly rebalanced portfolios. We form portfolios based on the weekly ML signals at the end of Tuesday and hold them until market close of the next Tuesday. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. The construction of the low-frequency ML signals is similar to that of the high-frequency ML , except that we use daily (weekly) stock returns to train different LASSO algorithms for predicting daily (weekly) ETF returns. $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). We multiply daily returns by 21 to get monthly returns in percentage. The row labeled “High-Low” reports the average monthly return spread between Quintiles 5 and 1, and the row labeled “CAPM Alpha” reports the monthly return spread adjusted by market exposure. The row labeled “Annual SR” reports the annualized Sharpe ratio for each long-short portfolio.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: Daily Rebalanced Portfolio Sorts</i>						
1 (Low)	0.36	0.16	0.25	0.03	0.12	0.01
2	0.36	0.34	0.57	0.40	0.40	0.30
3	0.54	0.64	0.63	0.64	0.64	0.74
4	0.59	0.68	0.77	0.77	0.74	0.83
5 (High)	0.90	0.78	0.95	1.04	0.98	1.34
High-Low	0.54	0.62	0.70	1.00	0.86	1.32
	(2.28)	(2.50)	(2.87)	(3.94)	(4.32)	(4.64)
CAPM Alpha	0.54	0.62	0.70	1.00	0.85	1.31
	(2.26)	(2.52)	(2.85)	(3.95)	(4.30)	(4.64)
Annual SR	0.73	0.84	0.94	1.31	1.58	1.74
<i>Panel B: Weekly Rebalanced Portfolio Sorts</i>						
1 (Low)	0.23	0.28	0.26	0.18	0.26	0.16
2	0.37	0.45	0.41	0.37	0.37	0.21
3	0.63	0.62	0.48	0.61	0.58	0.65
4	0.68	0.72	0.76	0.68	0.74	0.85
5 (High)	0.71	0.74	0.80	0.87	0.83	1.01
High-Low	0.43	0.41	0.49	0.63	0.57	0.85
	(2.00)	(2.14)	(2.38)	(2.52)	(2.78)	(3.22)
CAPM Alpha	0.42	0.41	0.48	0.64	0.56	0.85
	(1.90)	(2.14)	(2.37)	(2.62)	(2.77)	(3.22)
Annual SR	0.45	0.45	0.57	0.75	0.86	0.96

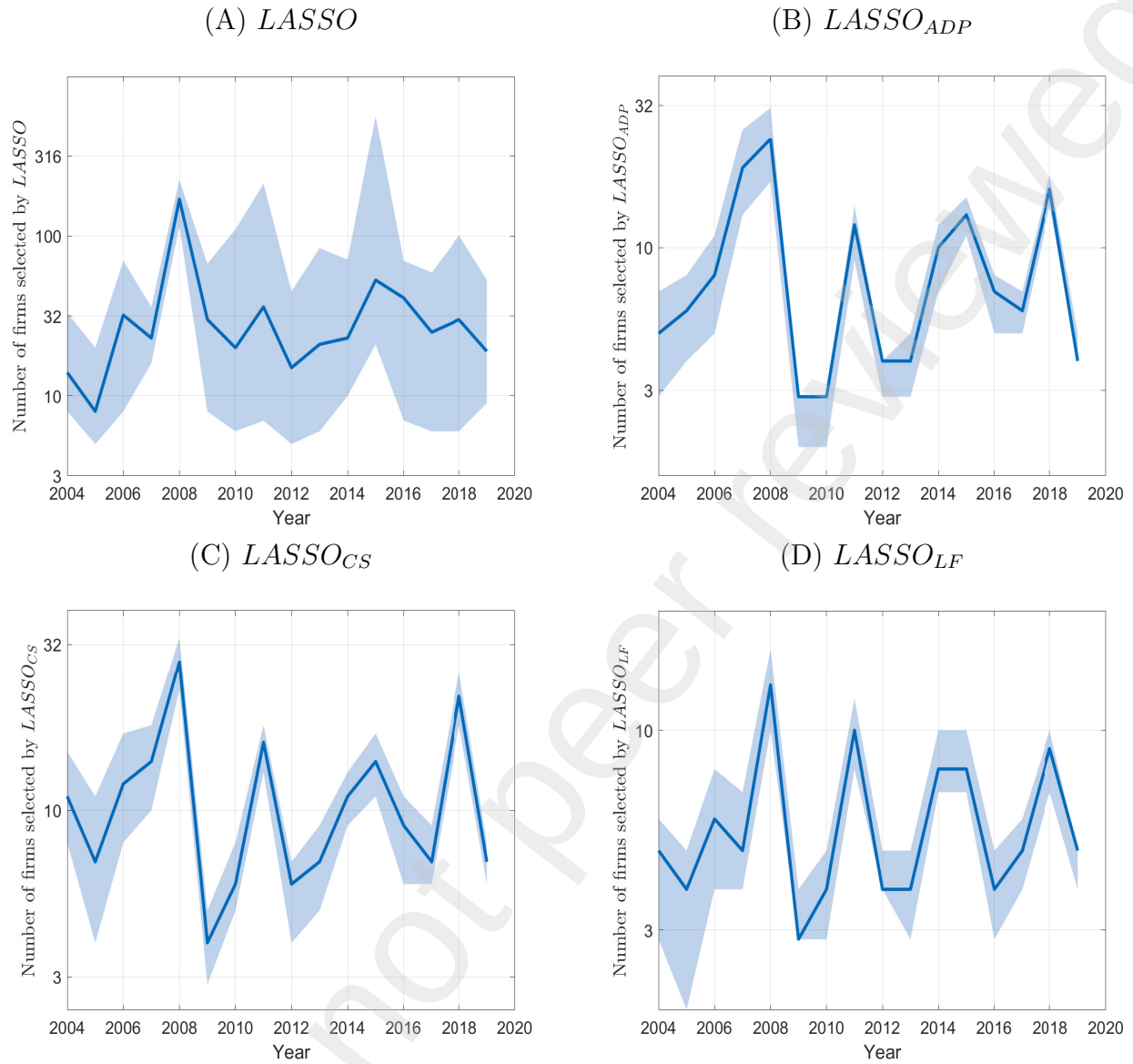


Fig. 1 Number of firms selected by LASSO: overnight signals

This figure plots the mean and 90% confidence interval of the number of firms selected by various LASSO methods to generate the overnight signal each year during our sample period. Specifically, $LASSO$ in Panel (A) is the simple LASSO signal, $LASSO_{ADP}$ in Panel (B) is the adaptive LASSO signal, $LASSO_{CS}$ in Panel (C) is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ in Panel (D) is the LASSO signal based on leading firms. All overnight LASSO signals are defined corresponding to $i = 1$ in Eq. (9).

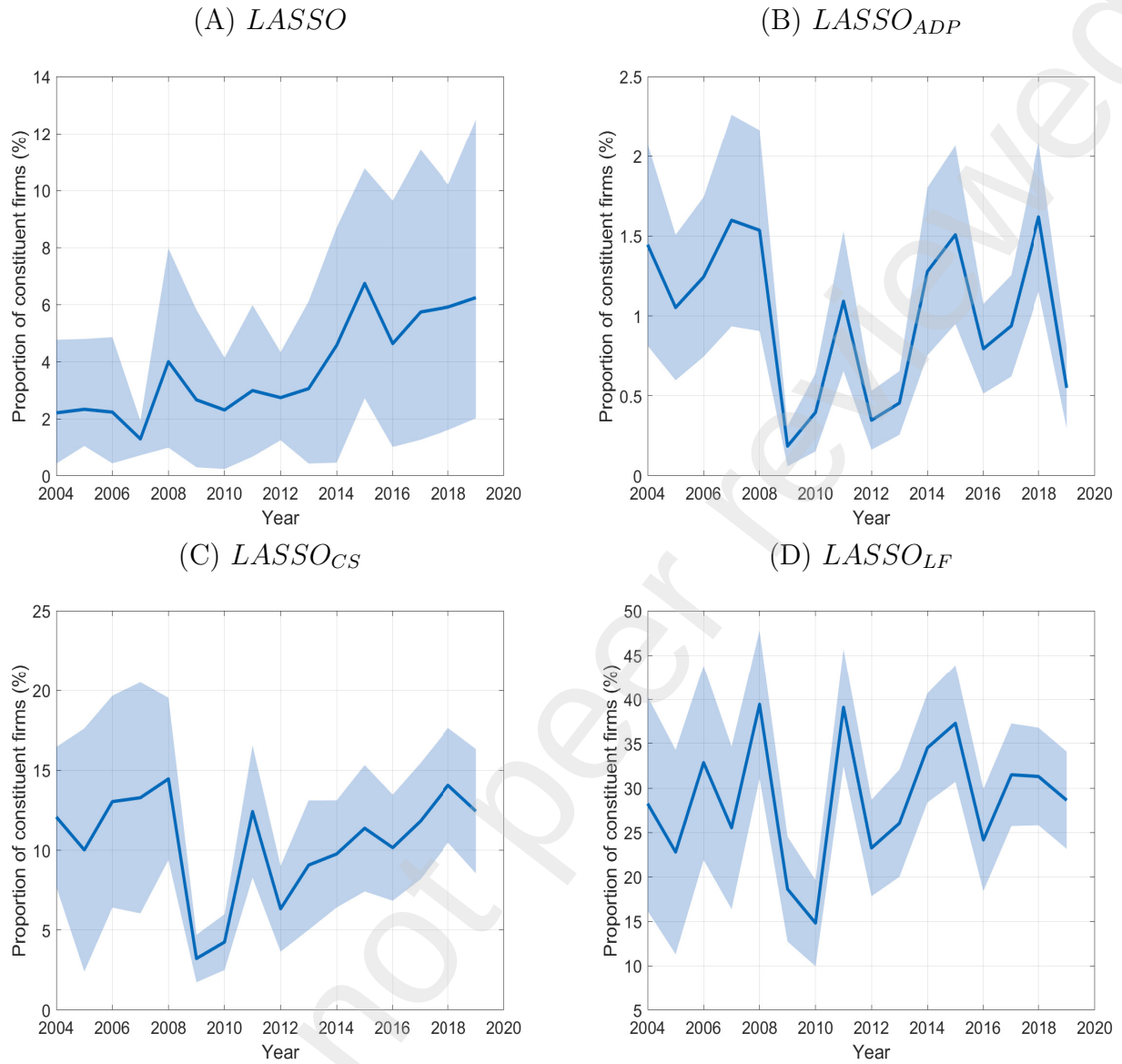


Fig. 2 Proportion of constituent firms selected by LASSO: overnight signals

This figure plots the mean and 90% confidence interval of the proportion of constituent firms selected by various LASSO methods to generate the overnight signal each year during our sample period. $LASSO$ in Panel (A) is the simple LASSO signal, $LASSO_{ADP}$ in Panel (B) is the adaptive LASSO signal, $LASSO_{CS}$ in Panel (C) is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ in Panel (D) is the LASSO signal based on leading firms. All overnight LASSO signals are defined corresponding to $i = 1$ in Eq. (9). To calculate the proportion of constituent firms for the $LASSO$ and $LASSO_{ADP}$ signals, the number of overlapping firms between LASSO-selected and ETF constituent firms is normalized by the total number of ETF constituents. For $LASSO_{CS}$ ($LASSO_{LF}$), the number of overlapping firms is normalized by the number of ETF constituent firms that are subset of the customer-supplier-linked (leading) firms.

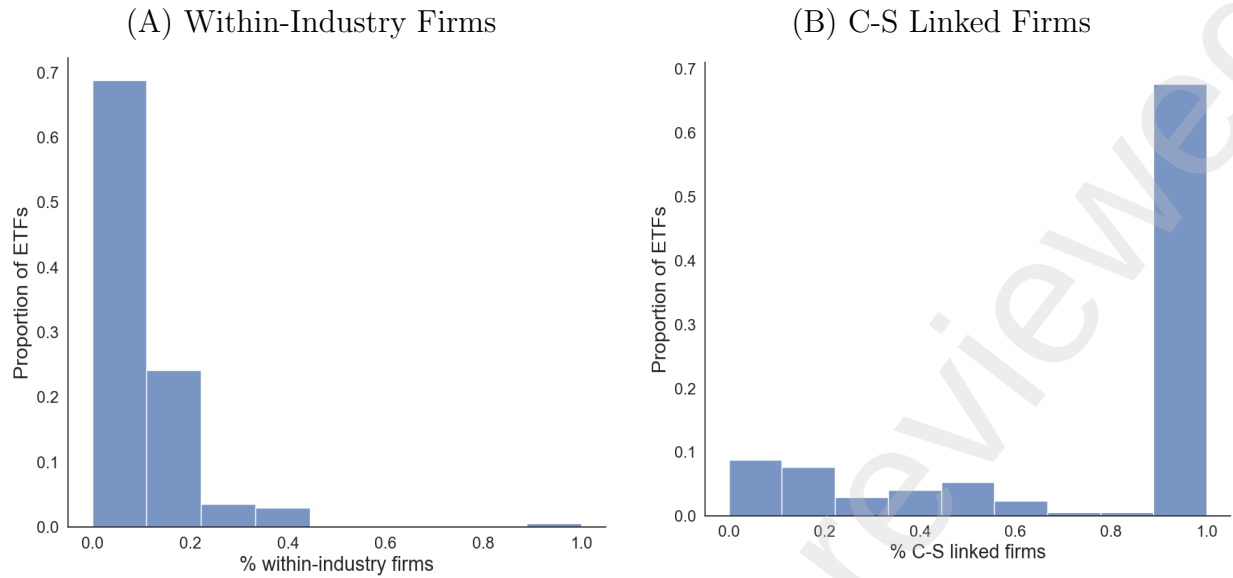
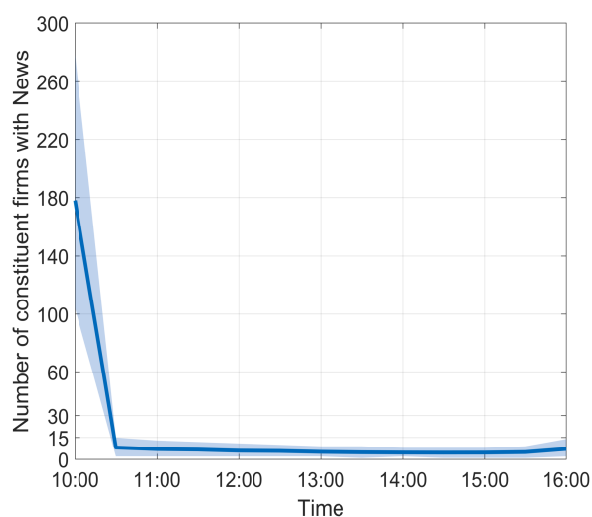


Fig. 3 Ratio of within-industry or C-S linked firms selected by LASSO: overnight signals

Panel A plots the distribution of the ratios of firms in the same industry of the ETF (i.e., within-industry firms) over LASSO-selected firms; Panel B plots the distribution of the ratios of the firms in customer-supplier industries of the ETF (i.e., C-S linked firms) over LASSO-selected firms. For each sector ETF and each LASSO training sample, we calculate the ratio of the within-industry firms among all LASSO-selected firms and the ratio of C-S linked firms among all LASSO-selected firms. Then, for each ETF, we compute the average of these two ratios over training samples. Lastly, we plot the histogram of the average ratios among ETFs (ratio for within-industry firms in Panel A and that for C-S linked firms in Panel B).

(A) Number of Constituent Firms with News



(B) Number of ETFs with News

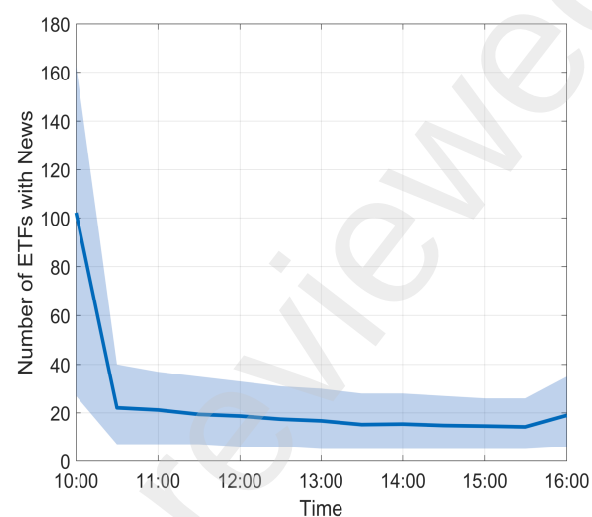


Fig. 4 Intraday news distributions

For each 30-minute interval, Panel (A) displays the average number of constituent firms with news and the corresponding 90% confidence interval; Panel (B) displays the average number of ETFs that have constituents with news and the 90% confidence interval. The x -axis denotes the ending time of each 30-minute interval, e.g., 10:00 denotes the overnight interval between 16:00 the previous day and 10:00, 10:30 denotes the interval between 10:00 and 10:30, etc.

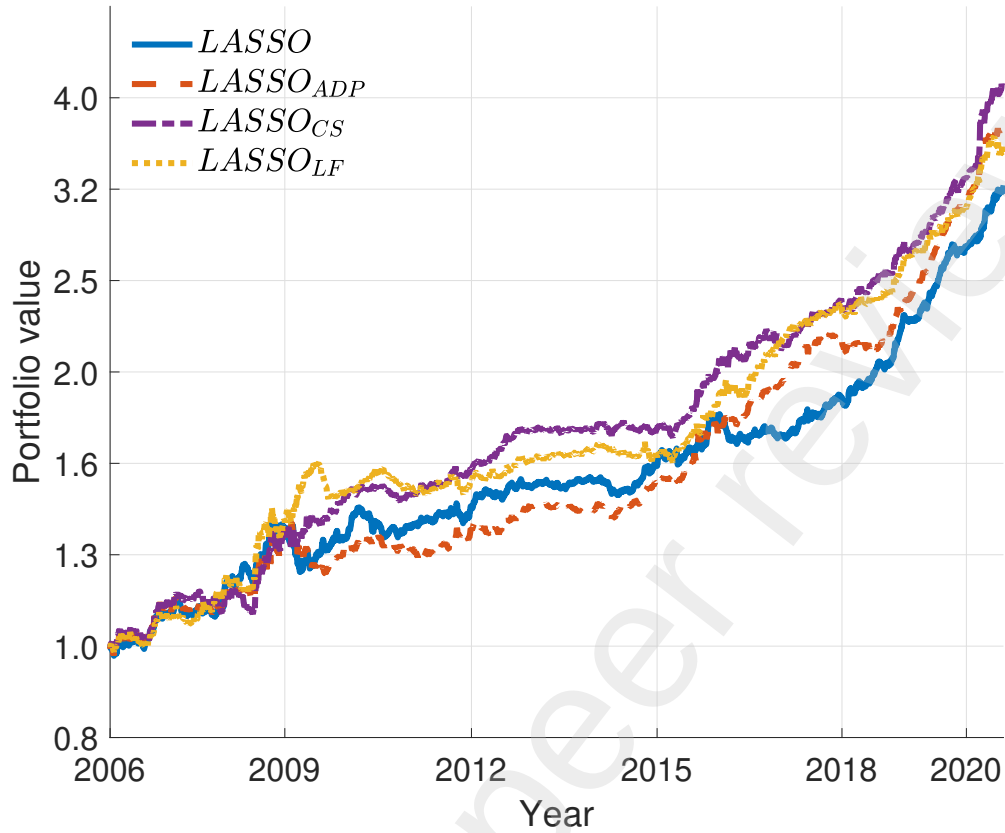


Fig. 5 Performance of trading strategies based on overnight LASSO signals

This figure shows the cumulative gains of trading strategies based on the overnight $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals. $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. All overnight LASSO signals are defined corresponding to $i = 1$ in Eq. (9). At 10:00 market open on each trading day t , we sort stocks into quintile portfolios based on each signal. Then we compute equal-weighted returns on each quintile portfolio with a holding period from 10:00 to 16:00 of the same day. Let $R_{high,d}$ and $R_{low,d}$ denote the returns on Quintile 5 and Quintile 1 on day d , respectively. The cumulative portfolio value based on the various overnight LASSO signals is computed as $W_d = W_{d-1}(1 + R_{high,d} - R_{low,d} + R_{rf,d})$, where $R_{rf,d}$ is the risk-free rate on day d and the initial investment is $W_1 = \$1$. Plotted is the time series of $\{W_d\}$. The scale in the figure is based on the logarithm with base ten.

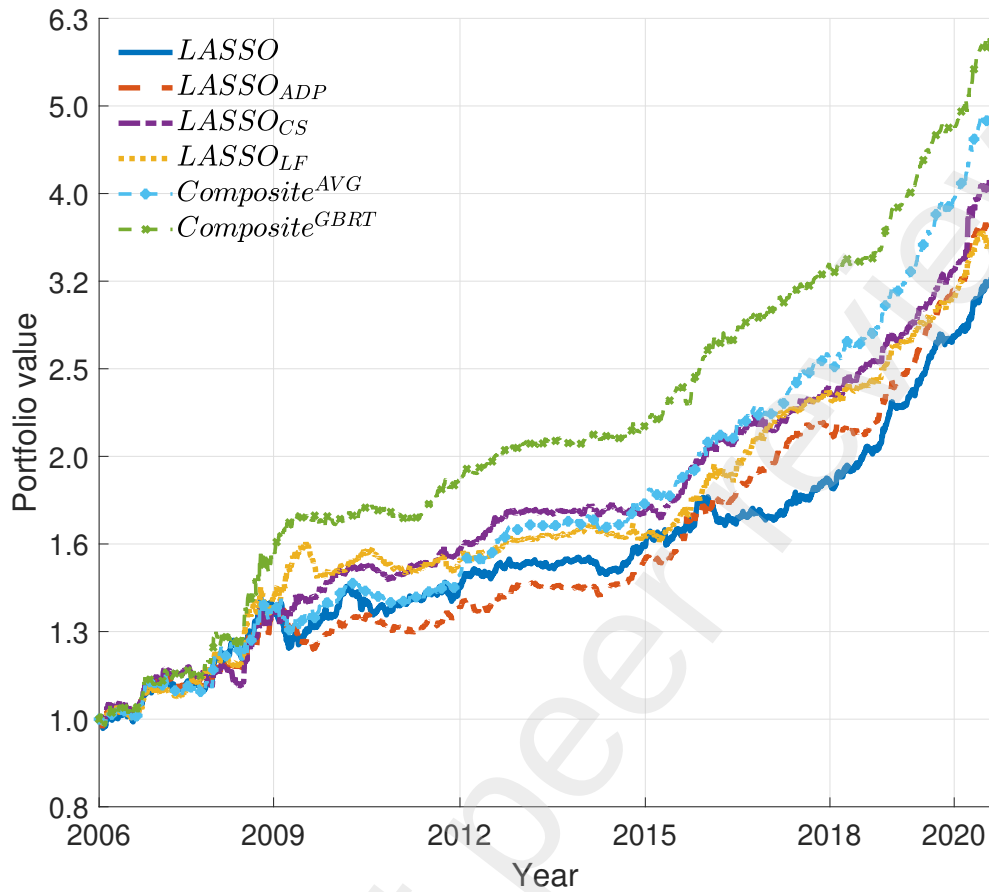


Fig. 6 Performance of trading strategies based on overnight LASSO and composite signals. This figure shows the cumulative gains of trading strategies based on the overnight $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms; all overnight LASSO signals are defined corresponding to $i = 1$ in Eq. (9). The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). At 10:00 market open on each trading day t , we sort stocks into quintile portfolios based on each signal. Then we compute equal-weighted returns on each quintile portfolio with a holding period from 10:00 to 16:00 of the same day. Let $R_{high,d}$ and $R_{low,d}$ denote the returns on Quintile 5 and Quintile 1 on day d , respectively. The cumulative portfolio value based on the various overnight LASSO signals is computed as $W_d = W_{d-1}(1 + R_{high,d} - R_{low,d} + R_{rf,d})$, where $R_{rf,d}$ is the risk-free rate on day d and the initial investment is $W_1 = \$1$. Plotted is the time series of $\{W_d\}$. The scale in the figure is based on the logarithm with base ten.

Internet Appendix

A.1. High-frequency Data Cleaning

We begin by removing entries that satisfy at least one of the following criteria: a price less than or equal to zero; a trade size less than or equal to zero; corrected trades, i.e., trades with Correction Indicator, CORR, other than 0, 1, or 2; and an abnormal sale condition, i.e., trades for which the Sale Condition, COND, has a letter code other than @, *, E, F, @E, @F, *E, and *F. We then assign a single value to each variable for each second. If one or multiple transactions have occurred in that second, we calculate the sum of volumes, the sum of trades, and the volume-weighted average price within that second. If no transaction has occurred in that second, we enter zero for volume and trades. For the volume-weighted average price, we use the entry from the nearest previous second. Motivated by our analysis of the trading volume distribution across different exchanges over time, we purposely incorporate information from all exchanges covered by the TAQ database.

A.2 Simulation Study

Our empirical results show that incorporating structural information into LASSO can help extract stronger signals and improve forecasting performance. In this section, we conduct a Monte Carlo simulation study to demonstrate the benefits of introducing structural priors to the standard LASSO problems for prediction.

We begin with simulating the return process for individual stocks r_i ($i = 1, \dots, N$) using a two-factor model:

$$\begin{aligned} r_{i,t} &= \bar{r}_i + \beta_i r_{m,t} + \eta_i r_{s,t} + \epsilon_{i,t}, \quad i = 1, \dots, N, \\ r_{m,t} &= \bar{r}_m + \epsilon_{m,t}, \\ r_{s,t} &= \rho \bar{r}_s + (1 - \rho) r_{s,t-1} + \epsilon_{s,t}, \end{aligned} \tag{22}$$

where $r_{m,t}$ denotes the market factor and $r_{s,t}$ denotes the sector factor; $\epsilon_{i,t} \sim \mathcal{N}(0, 0.01^2)$, $\epsilon_{s,t} \sim \mathcal{N}(0, 0.006^2)$, and $\epsilon_{m,t} \sim \mathcal{N}(0, 0.003^2)$. To simplify the model, we assume all return processes are mean-zero (e.g., $\bar{r}_i = \bar{r}_m = \bar{r}_s = 0$), the market exposure $\beta_i = 1$ for all stocks, and $\rho = 0.5$. However, we allow the sector exposure η_i to vary across stocks. Specifically, in each Monte Carlo replication, we simulate 30-minute returns for 3,000 stocks over 15 years (# obs. per stock = $13 \times 252 \times 15$). Out of the 3,000 stocks, 2,500 stocks have low $\eta_i \sim U(0, 0.05)$, 400 stocks have medium $\eta_i \sim U(0.1, 0.4)$, and 100 stocks have high $\eta_i \sim U(0.8, 1.0)$.

After generating the full time series of 3,000 stocks, we create 100 synthetic “ETFs” each as equal-weighted average of 50 random stocks. Given that sector ETFs tilt towards large stocks with high sector exposure, for each ETF, we draw 70% of its constituents from a universe with $\eta_i \sim U(0.1, 0.4)$ or $U(0.8, 1)$, and 30% constituents from a universe with $\eta_i \sim U(0, 0.05)$. Due to randomness, the resulting ETFs will have different levels of sector exposure.

With the simulated data on individual stocks and ETFs, we are able to train different LASSO signals to predict 30-minute ahead ETF returns using lagged stock returns. The four signals we consider include: 1) *LASSO*, which is the simple LASSO signal using all 3,000 firms as predictors; 2) *LASSO_{ADP}*, which is the adaptive LASSO signal using all 3,000 firms as predictors; 3) *LASSO_{TOP500}*, which is the LASSO signal using top 500 firms with medium/high sector exposure as predictors (e.g., $\eta_i \sim U(0.1, 0.4)$ or $U(0.8, 1)$); 4) *LASSO_{TOP100}*, which is the LASSO signal using top 100 firms with high sector exposure as predictors (e.g., $\eta_i \sim U(0.8, 1.0)$). We repeat the previous steps 100 times to generate 100 Monte Carlo samples. For each Monte Carlo sample, we divide the 15-year time series into 3 consecutive sub-samples consisting of 4-year training, 1-year validation, and 10-year testing. Then we estimate each of the four LASSO models for a given shrinking parameter λ using the training sample, choose the optimal λ using the validation sample, and construct LASSO signals to predict ETF returns using the testing sample.

We are interested in formally assessing whether different LASSO signals generate significantly different economic gains. To do so, for Monte Carlo sample n and LASSO signal i , we run Fama and MacBeth (1973) cross-sectional regressions and record the signal coefficient $\gamma_{LASSO,i,n}$. The resulting economic gain, or the rise in annual ETF returns predicted by a two-standard-deviation increase in the LASSO signal, is given by $Gain_{i,n} = 2 \times \gamma_{LASSO,i,n} \times SD_{LASSO,i,n} \times 252$, where $SD_{LASSO,i,n}$ is the average cross-sectional standard deviation of the LASSO signal. To compare the economic gain between two LASSO signals i and j , we denote the gain differential from sample n as $d_{i,j,n} = Gain_{i,n} - Gain_{j,n}$. The test statistic for pairwise comparison of the two signals is $z_{i,j} = \overline{d_{i,j}} / \hat{\sigma}(d_{i,j})$, where $\overline{d_{i,j}}$ and $\hat{\sigma}(d_{i,j})$ are the mean and Newey and West (1987) standard error of $d_{i,j,n}$ over different Monte Carlo samples.

The table below reports the test t -statistics for pairwise comparisons of economic gains for a model in the column versus a model in the row. A positive t -statistic indicates that the column model outperforms the row model; *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Among the four LASSO signals, $LASSO_{TOP100}$ ($LASSO$) significantly outperforms (underperforms) the rest; $LASSO_{Top500}$ and $LASSO_{ADP}$ are in between. The results are consistent with the empirical analyses that LASSO signals based on screened feature set with stronger predictors beat simple LASSO signals using the entire stock universe.

<i>Test Statistics for Signal Comparison</i>			
	$LASSO_{ADP}$	$LASSO_{Top500}$	$LASSO_{Top100}$
$LASSO$	3.92***	4.13***	6.53***
$LASSO_{ADP}$		0.79	3.66***
$LASSO_{Top500}$			2.81***

To further compare the four LASSO signals, we form long-short portfolios and record portfolio returns. Fig. A.2 shows the average cumulative gains over 100 Monte Carlo samples from trading strategies based on each of the four LASSO signals. Consistent with the results in the above table, $LASSO_{TOP100}$ signal yields the highest cumulative gains, followed by $LASSO_{TOP500}$ and $LASSO_{ADP}$, while $LASSO$ performs the worst. In a nutshell, the

simulation study well supports the empirical finding that pure statistical algorithms such as LASSO and adaptive LASSO do not perform as well as algorithms incorporating priori knowledge about the feature set.

A.3 Additional Robustness Checks

A.3.1 NAV-based ETF Return Predictability

Due to market frictions, ETF price may deviate from the underlying portfolio value, resulting in ETF mispricing. Does ETF mispricing have any effect on the documented ETF predictability? To provide further insights, we first assess the degree of mispricing in our sector ETF sample by estimating the average percentage deviation of close prices from daily NAV for a given ETF as follows:

$$Deviation_j = \frac{1}{D_j} \sum_d \left| \frac{P_{j,d} - NAV_{j,d}}{NAV_{j,d}} \right|, \quad (23)$$

where $P_{j,d}$ is the close price of ETF j on day d , $NAV_{j,d}$ is the end-of-day net asset value, and D_j is the number of daily observations for ETF j . After estimating the percentage price deviation from NAV for each ETF, we plot the histogram in Fig. A.3. The percentage deviation ranges from around 0.03% to around 0.45%, with mean and median around 0.1%, implying that prices of sector ETFs closely track the net asset value of the funds on average.

To directly examine how ETF mispricing may impact ETF predictability, we estimate daily ETF returns based on end-of-day NAV from the CRSP Mutual Fund Database. Then we retrain four NAV-based LASSO signals using daily stock returns to predict daily NAV-based ETF returns. Finally, we construct two composite signals via simple averaging or GBRT. Table A.5 summarizes the Fama-MacBeth regression and portfolio sorting results using the retrained signals to predict NAV-based ETF returns. Overall, the results are highly similar to those based on close prices. Take $Composite^{GBRT}$ for example. The coefficient and t -statistic

of this signal are now 0.271 and 4.24, compared to the coefficient of 0.29 and t -statistic of 4.23 based on close prices from Table 9. The monthly return spread and annual Sharpe ratio generated by *Composite*^{GBRT} are 1.29% and 1.64, similar to the spread of 1.31% and the Sharpe ratio of 1.74 based on close prices from Panel A of Table 10. All in all, the results suggest that the documented ETF predictability is not affected by ETF mispricing.

A.3.2 Predicting ETF Returns with Elastic Net Method

We have been focusing on using LASSO to aggregate individual firms' information for predicting future ETF returns. The LASSO is capable of selecting sparse features by forcing the coefficients of unimportant predictors to be zero. One concern is that the LASSO struggles with colinear features, in which it will select only one predictor from a group of highly correlated features and ignore the others. To solve this problem, we apply the Elastic Net method which includes both $L1$ and $L2$ norm regularization terms:

$$\hat{\beta}_{j,i}^{Enet} = \underset{\beta_{j,i}}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_d \left(r_{j,d,i+1}^{etf} - \beta_{j,i,0} - \sum_{s=1}^p \beta_{j,i,s} r_{s,d,i}^{stc} \right)^2 + \lambda_{j,i} \sum_{s=1}^p |\beta_{j,i,s}| + \lambda'_{j,i} \sum_{s=1}^p \beta_{j,i,s}^2 \right\}, \quad (24)$$

where λ' is the hyperparameter that controls the amount of shrinkage based on $L2$ penalty. Table A.6 reports the Fama-MacBeth regression results predicting each 30-minute ETF returns using *ElasticNet*; the last column presents the average coefficients, t -statistics, and adjusted R^2 across different 30-minute intervals. The average coefficient of *ElasticNet* is 0.169 with a t -statistic of 10.59, and the average adjusted- R^2 is 14.92%. Compared with Table 2, *ElasticNet* exhibits predictability similar to the LASSO method. Thus, our main results are robust to the alternative linear machine learning method.

A.3.3 Seasonality in ETF Return Predictability

In this subsection, we examine the seasonality in ETF return predictability, namely the day-of-week effects. Table A.7 summarizes the results for the long-short portfolio performance on each day of the week. Panel A reports the average portfolio returns, and Panel B reports the corresponding annualized Sharpe ratios. The returns to the long-short portfolios formed on the basis of different LASSO signals remain statistically significant and economically large across all days of the week: the t -statistics of the average portfolio returns are all above two and the annual Sharpe ratios are over 0.9 across the board. In addition, the predictive power of the signals remain in the order of $(LASSO, LASSO_{ADP}) < (LASSO_{CS}, LASSO_{LF}) < (Composite^{AVG}, Composite^{GBRT})$ on most days of the week. Take Friday as an example. $LASSO$ and $LASSO_{ADP}$ both generate an average return spread around 0.65% per month; the corresponding number for $LASSO_{CS}$ and $LASSO_{LF}$ is around 0.73%; for portfolios formed by the two composite signals, their average return spreads are 0.73% and 1.13%.

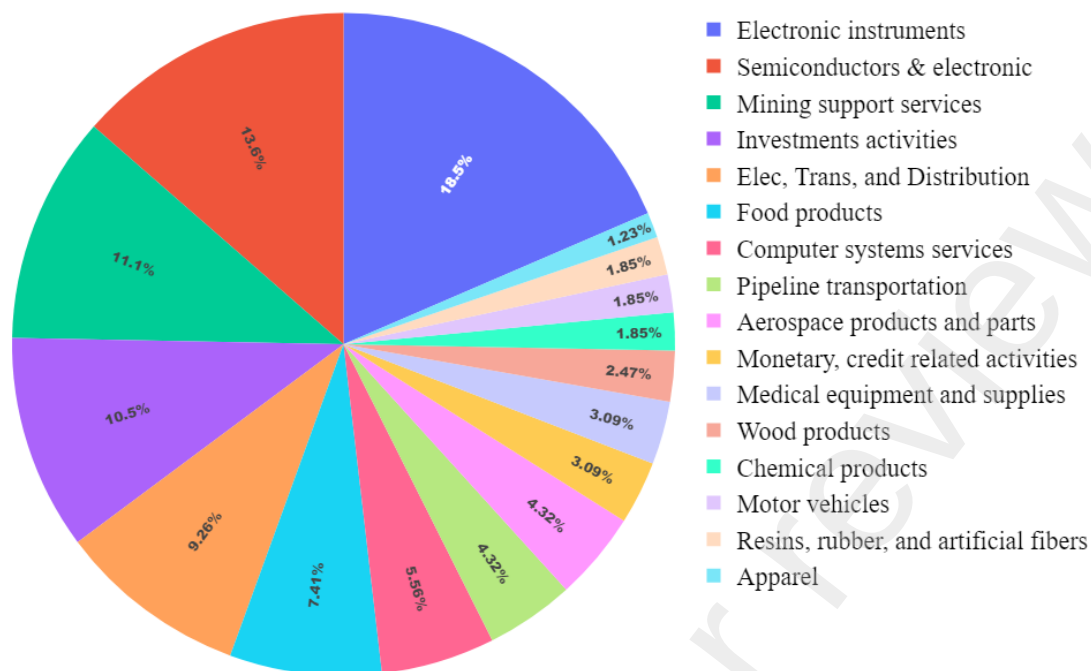


Fig. A.1 Composition of sector ETFs by industry

This figure plots the snapshot of the percentage sector ETFs by industry at the end of the sample period in July 2020; only the top 16 industries with at least 1% ETFs are displayed.

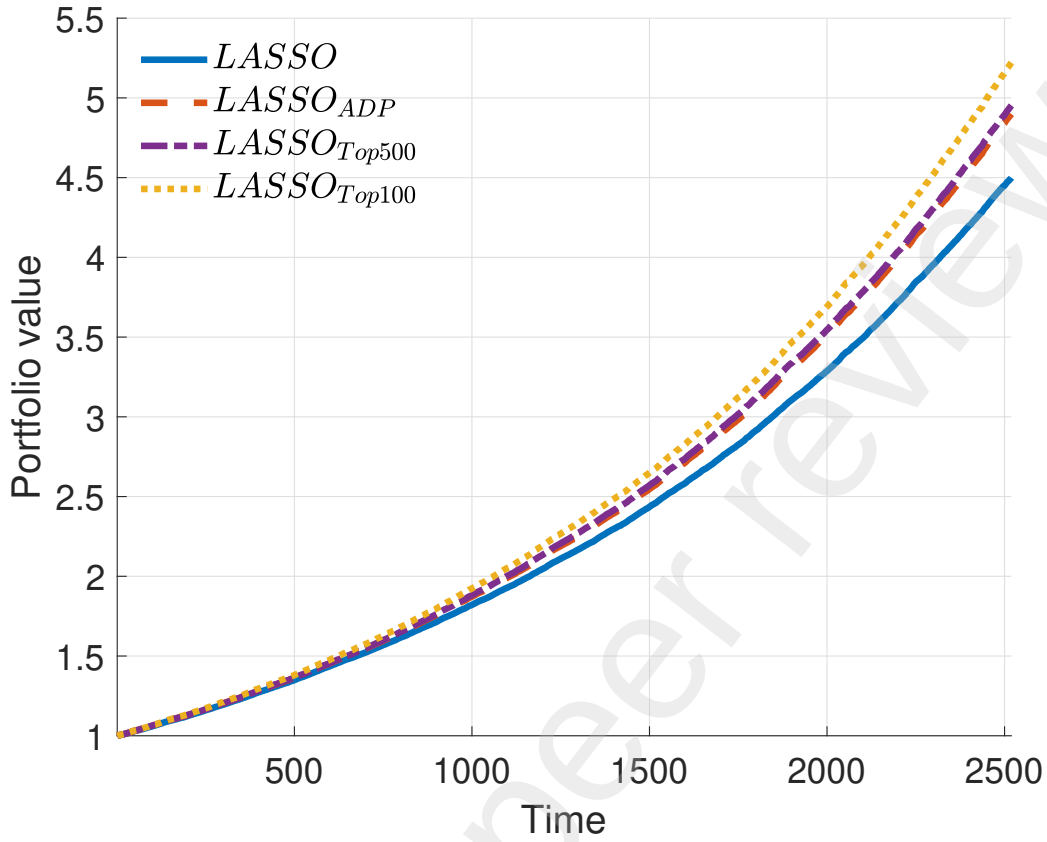


Fig. A.2 Simulation study: average cumulative gains of trading strategies

This figure shows the average cumulative gains of trading strategies based on the $LASSO$, $LASSO_{ADP}$, $LASSO_{TOP500}$, and $LASSO_{TOP100}$ signals over a 10-year testing sample. $LASSO$ is the simple LASSO model using all 3,000 firms as predictors. $LASSO_{ADP}$ denotes the Adaptive LASSO using all 3,000 firms as predictors. $LASSO_{TOP500}$ is LASSO model using 500 firms with medium/high sector exposure as predictors ($\eta_i \in (0.8, 1.0)$ or $(0.1, 0.4)$). $LASSO_{TOP100}$ stands for LASSO model using 100 firms with high sector exposure as predictors ($\eta_i \in (0.8, 1.0)$). Let $R_{high,d}$ and $R_{low,d}$ denote the returns on Quintile 5 and Quintile 1 on day d , respectively. The cumulative portfolio value based on the LASSO signals is computed as $W_d = W_{d-1}(1 + R_{high,d} - R_{low,d})$, where the initial investment is $W_1 = \$1$. We compute the average of W_d cross 100 Monte Carlo samples.

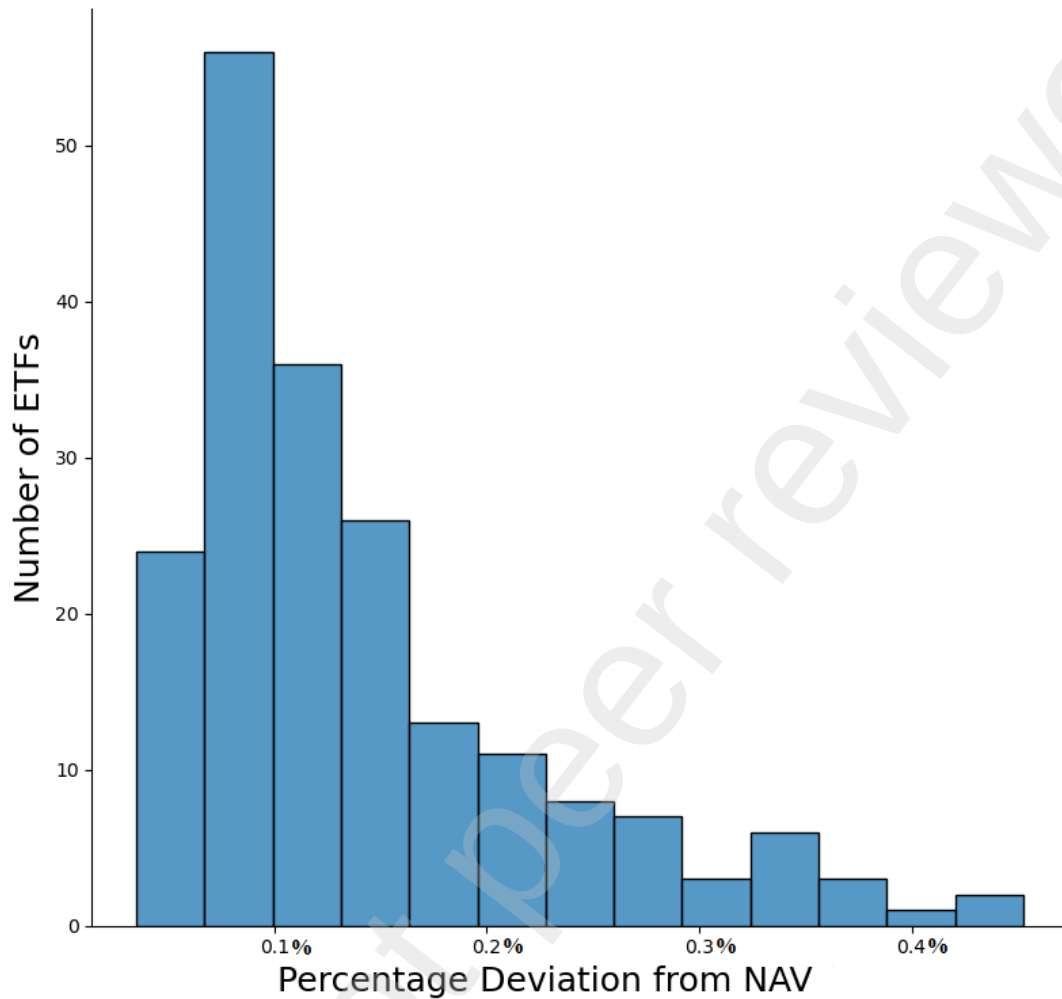


Fig. A.3 Percentage deviation of ETF close prices from NAV

This figure plots the histogram of the average percentage deviation of close prices from NAV for the sector ETFs in our sample. For each ETF, the average percentage deviation from NAV is calculated according to Eq. (23).

Table A.1 Controlling for the intra-industry lead-lag effect of Hou (2007)

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting daily or weekly ETF returns by the corresponding machine-learning (ML) signals learned from lagged daily or weekly stock returns, controlling for the intra-industry lead-lag effect of Hou (2007), i.e., the lead-lag effect between big firms and small firms is predominantly an intra-industry phenomenon. Both daily and weekly (Tuesday-to-Tuesday) returns are calculated using the close prices. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. The construction of the low-frequency ML signals is similar to that of the high-frequency signals, except that we use daily (weekly) stock returns to train different LASSO algorithms for predicting daily (weekly) ETF returns. ML signals include $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). $News$ is the lagged daily (Panel A) or weekly (Panel B) news return aggregated from overnight and intraday 30-minute news returns. Big is the equal-weighted average return of top size-decile firms within the same industry as the sector ETF. All regressions control for $ILLQ$, $RVOL$, Mom , and $Size$, all of which are defined in Table 9.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: Predicting Daily ETF Returns</i>						
Intercept	7.532 (2.71)	6.930 (2.75)	6.775 (2.38)	6.718 (2.28)	6.763 (2.32)	6.521 (2.11)
ML	0.070 (2.40)	0.106 (2.98)	0.180 (3.26)	0.255 (3.88)	0.226 (3.94)	0.289 (4.23)
$News$	0.002 (2.32)	0.004 (2.28)	0.002 (2.12)	0.003 (2.51)	0.003 (2.50)	0.002 (2.13)
Big	0.014 (1.90)	0.013 (1.87)	0.013 (1.83)	0.013 (1.86)	0.013 (1.88)	0.012 (1.82)
ILLIQ	0.369 (1.58)	0.404 (1.58)	0.387 (1.67)	0.327 (1.44)	0.322 (1.42)	0.337 (1.47)
RVOL	-8.818 (-1.59)	-6.813 (-1.23)	-7.612 (-1.30)	-9.550 (-1.70)	-6.216 (-1.15)	-5.731 (-1.05)
Mom	2.951 (1.60)	2.822 (1.44)	3.328 (1.74)	2.261 (1.21)	2.792 (1.46)	2.634 (1.36)
Size	0.199 (0.41)	0.461 (0.93)	0.612 (1.18)	0.264 (0.55)	0.195 (0.44)	0.218 (0.45)
Adj- R^2 (%)	21.31	21.39	21.53	21.94	22.21	22.67
<i>Panel B: Predicting Weekly ETF Returns</i>						
Intercept	31.872 (2.41)	31.437 (2.81)	30.123 (2.76)	30.246 (2.48)	31.154 (2.56)	30.134 (2.52)
ML	0.082 (1.88)	0.085 (2.15)	0.122 (2.26)	0.136 (2.57)	0.133 (2.69)	0.151 (2.80)
$News$	0.001 (0.07)	0.001 (0.09)	0.000 (0.01)	0.002 (0.27)	0.001 (0.07)	0.001 (0.07)
Big	0.029 (1.75)	0.031 (1.86)	0.033 (2.03)	0.029 (1.81)	0.031 (1.86)	0.029 (1.80)
ILLIQ	1.080 (1.07)	1.582 (1.66)	1.441 (1.58)	1.166 (1.29)	0.852 (0.94)	0.825 (0.88)
RVOL	-57.149 (-2.68)	-52.674 (-2.45)	-53.617 (-2.34)	-47.233 (-2.10)	-45.069 (-2.20)	-44.393 (-2.10)
Mom	12.088 (1.05)	14.903 (1.28)	15.958 (1.33)	12.845 (1.08)	10.059 (0.92)	10.122 (0.97)
Size	2.641 (1.27)	3.664 (1.76)	3.154 (1.62)	2.709 (1.42)	2.323 (1.23)	2.454 (1.24)
Adj- R^2 (%)	20.15	20.48	20.75	21.32	21.54	22.17

Table A.2 Predicting Fama-French 49 industry portfolios

Panel A reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MacBeth cross-sectional regressions predicting daily returns of 49 industry portfolios by machine-learning (ML) signals learned from lagged daily stock returns. Daily equal-weighted industry portfolio returns are obtained from Ken French's Data Library; daily stock returns are calculated using close prices. The construction of the industry portfolio ML signals is similar to that of the sector ETF signals, except that we use daily stock returns to train different LASSO algorithms for predicting daily industry portfolio returns. ML signals include $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. $LASSO$ is the raw LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). Since the analyst coverage data used to construct $LASSO_{LF}$ became available in 1976 and we require five-year data for training and validation, the out-of-sample period for the industry portfolio analyses is from January 1981 to July 2020. All regressions control for RET , $ILLQ$, $RVOL$, Mom , and $Size$, all of which are defined in Table 9. RET , $RVOL$, and Mom are calculated based on daily industry portfolio returns, while industry $ILLQ$ and $Size$ are constructed by equally averaging the corresponding measure of stocks in the same industry. Panel B reports the performance of quintile portfolios formed on the basis of daily machine-learning (ML) signals over the same out-of-sample period. At the end of each day d , we sort industries into quintile portfolios based on ML signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys industries in the top quintile with high signal values and sells industries in the bottom quintile with low signal values, with a holding period from market close on day d until market close of the next day. We multiply daily returns by 21 to get monthly returns in percentage. The row labeled "High-Low" reports the average monthly return spread between Quintiles 5 and 1, and the row labeled "CAPM Alpha" reports the monthly return spread adjusted by market exposure. The row labeled "Annual SR" reports the annualized Sharpe ratio for each long-short portfolio.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: Fama-MacBeth Regressions</i>						
Intercept	0.001 (2.00)	0.001 (1.82)	0.001 (1.68)	0.001 (2.03)	0.001 (2.01)	0.001 (2.02)
ML	0.111 (4.74)	0.116 (5.05)	0.144 (5.76)	0.161 (6.01)	0.146 (6.42)	0.172 (7.12)
RET	0.022 (7.58)	0.023 (8.02)	0.023 (8.09)	0.022 (7.74)	0.021 (7.71)	0.020 (7.67)
$ILLIQ$	-0.689 (-1.10)	-0.859 (-1.37)	-0.861 (-1.37)	-1.016 (-1.61)	-1.083 (-1.69)	-1.006 (-1.41)
$RVOL$	-24.025 (-1.18)	-22.186 (-1.09)	-17.353 (-0.85)	-22.125 (-1.09)	-21.125 (-1.02)	-22.013 (-1.07)
Mom	7.558 (3.74)	7.288 (3.60)	6.873 (3.36)	7.740 (3.83)	7.096 (3.53)	7.746 (3.88)
$Size$	-1.529 (-1.56)	-1.671 (-1.71)	-1.640 (-1.67)	-1.960 (-1.99)	-1.760 (-1.82)	-1.685 (-1.76)
Adj- R^2 (%)	14.01	14.02	14.29	14.47	14.41	15.21
<i>Panel B: Portfolio Sorts</i>						
1 (Low)	0.54	0.46	0.39	0.36	0.44	0.27
2	0.85	0.79	0.90	0.82	0.84	0.82
3	1.11	1.05	1.10	1.02	1.07	1.12
4	1.41	1.38	1.31	1.45	1.39	1.47
5 (High)	1.60	1.80	1.82	1.86	1.81	1.89
High-Low	1.07 (7.87)	1.34 (10.27)	1.43 (10.74)	1.49 (11.06)	1.37 (11.36)	1.62 (12.06)
CAPM Alpha	1.06 (7.86)	1.33 (10.23)	1.41 (10.72)	1.49 (11.05)	1.35 (11.32)	1.60 (12.01)
Annual SR	1.30	1.67	1.79	1.87	2.01	2.21

Table A.3 Predicting 30-minute ahead ETF returns: high-frequency LASSO signal trained from residual returns

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting every intraday half-hour ETF return in interval i ($i = 2, \dots, 13$) on day d by the $LASSO_{RES}$ and $News$ signals calculated using inputs from interval $i - 1$, where $LASSO_{RES}$ is the simple LASSO signal trained from firm residual returns defined in Eq. (19). Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. $News$ is the news-driven signal from Eq. (10). RET is the lagged overnight or 30-minute ETF return. $ILLQ$ is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. $RVOL$ is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. Mom for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. $Size$ is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return Interval													
Start	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	Mean
End	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	
Intercept	-0.006 (-0.06)	0.123 (1.25)	0.148 (1.76)	0.035 (0.48)	0.064 (0.89)	0.065 (0.91)	-0.049 (-0.68)	-0.032 (-0.46)	-0.105 (-1.32)	-0.174 (-1.12)	0.149 (1.92)	-0.149 (-1.57)	0.006 (0.17)
$LASSO_{RES}$	0.112 (7.47)	0.188 (10.32)	0.200 (12.93)	0.158 (11.04)	0.209 (13.59)	0.223 (14.56)	0.199 (12.05)	0.184 (10.96)	0.156 (9.36)	0.148 (11.15)	0.145 (7.73)	0.124 (7.41)	0.170 (10.71)
$News$	0.020 (21.96)	0.106 (1.57)	-0.056 (-0.70)	0.072 (1.57)	0.019 (0.81)	0.013 (1.28)	0.048 (1.19)	-0.069 (-1.35)	0.107 (1.02)	0.217 (1.08)	0.009 (0.69)	0.520 (0.86)	0.084 (2.50)
RET	-0.079 (-26.91)	-0.087 (-21.00)	-0.103 (-25.04)	-0.104 (-23.85)	-0.119 (-28.13)	-0.130 (-32.73)	-0.138 (-30.54)	-0.147 (-31.53)	-0.161 (-31.52)	-0.157 (-35.68)	-0.164 (-35.85)	-0.207 (-40.32)	-0.133 (-30.26)
ILLIQ	0.001 (0.01)	0.046 (0.73)	0.019 (0.34)	-0.018 (-0.37)	0.033 (0.68)	0.093 (1.97)	-0.028 (-0.55)	-0.010 (-0.22)	-0.037 (-0.73)	-0.134 (-1.23)	0.028 (0.48)	-0.054 (-0.84)	-0.005 (0.02)
RVOL	-1.304 (-0.90)	-0.099 (-0.07)	-2.266 (-1.90)	-1.679 (-1.66)	-0.368 (-0.37)	0.109 (0.12)	-1.675 (-1.77)	-1.868 (-2.11)	-0.453 (-0.45)	0.872 (0.93)	0.111 (0.13)	1.336 (1.37)	-0.607 (-0.56)
Mom	0.370 (0.85)	0.084 (0.22)	-0.116 (-0.37)	0.087 (0.29)	-0.571 (-2.11)	0.499 (1.79)	-0.510 (-2.04)	-0.028 (-0.12)	0.194 (0.75)	0.374 (1.53)	-0.182 (-0.72)	0.161 (0.53)	0.030 (0.05)
Size	0.003 (0.02)	-0.107 (-0.95)	-0.098 (-0.97)	-0.089 (-0.96)	0.002 (0.03)	0.183 (2.25)	0.005 (0.06)	0.002 (0.02)	0.000 (0.00)	-0.135 (-0.96)	-0.101 (-1.02)	-0.005 (-0.04)	-0.028 (-0.21)
Adj- R^2 (%)	15.75	15.07	15.10	14.89	14.21	13.79	14.37	14.70	15.21	15.76	15.72	14.73	14.94

Table A.4 Performance of trading strategies after transaction costs

This table reports the trading cost-adjusted performance of strategies formed on the basis of different overnight LASSO signals over the entire out-of-sample period between January 2006 and July 2020. At 10:00 of each trading day d , following the rules in Section 6.4, we form portfolios based on their overnight $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All overnight LASSO signals correspond to interval $i = 1$ in Eq. (9). $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). We multiply daily returns by 21 to get monthly returns in percentage. The row labeled “Annual SR” reports the annualized Sharpe ratio for each portfolio.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
Return	0.31 (2.15)	0.31 (2.23)	0.42 (2.23)	0.40 (2.39)	0.41 (3.00)	0.64 (3.47)
Annual SR	0.59	0.67	0.84	0.82	1.13	1.30

Table A.5 Predicting NAV-based ETF returns

Panel A reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MacBeth cross-sectional regressions predicting daily NAV-based ETF returns by machine-learning (ML) signals learned from lagged daily stock returns. Daily NAV-based ETF returns are calculated using the end-of-day net asset value (NAV) of the fund obtained from the CRSP Mutual Fund Database; daily stock returns are calculated using close prices. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. The construction of the NAV-based ML signals is similar to that of the close-price-based signals, except that we use daily stock returns to train different LASSO algorithms for predicting daily NAV-based ETF returns. ML signals include $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. $LASSO$ is the raw LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). $News$ is the daily news return aggregated from overnight and intraday 30-minute news returns. All regressions control for RET , $ILLQ$, $RVOL$, Mom , and $Size$, all of which are defined in Table 9. Panel B reports the performance of quintile portfolios formed on the basis of daily NAV-based machine-learning (ML) signals. At the market close of each trading day d , we sort ETFs into quintile portfolios based on NAV-based ML signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom quintile with low signal values, with a holding period from market close on day d until market close of the next day. We multiply daily returns by 21 to get monthly returns in percentage. The row labeled “High-Low” reports the average monthly return spread between Quintiles 5 and 1, and the row labeled “CAPM Alpha” reports the monthly return spread adjusted by market exposure. The row labeled “Annual SR” reports the annualized Sharpe ratio for each long-short portfolio.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: Fama-MacBeth Regressions</i>						
Intercept	5.762 (2.28)	4.576 (1.81)	6.014 (2.34)	4.575 (1.78)	5.039 (2.05)	4.546 (1.58)
ML	0.074 (2.65)	0.154 (2.91)	0.223 (3.54)	0.253 (3.79)	0.219 (3.85)	0.271 (4.24)
$News$	0.006 (3.33)	0.005 (2.91)	0.005 (2.84)	0.006 (3.79)	0.006 (2.74)	0.006 (3.54)
RET	-0.013 (-2.01)	-0.010 (-1.50)	-0.012 (-1.82)	-0.012 (-1.90)	-0.013 (-1.92)	-0.013 (-1.94)
$ILLIQ$	0.211 (0.99)	0.213 (1.02)	0.233 (1.13)	0.228 (1.04)	0.209 (1.01)	0.212 (1.02)
$RVOL$	-9.951 (-1.95)	-9.938 (-1.93)	-7.726 (-1.49)	-10.460 (-1.99)	-8.726 (-1.59)	-9.447 (-1.69)
Mom	1.525 (0.98)	1.064 (0.67)	1.067 (0.67)	0.509 (0.33)	1.569 (0.65)	0.793 (0.43)
$Size$	0.267 (0.48)	0.454 (0.66)	0.150 (0.20)	0.278 (0.61)	0.170 (0.21)	0.196 (0.41)
Adj- R^2 (%)	24.29	24.34	24.37	24.49	24.35	24.61
<i>Panel B: Portfolio Sorts</i>						
1 (Low)	0.35	0.34	0.29	0.21	0.24	0.13
2	0.62	0.88	0.31	0.32	0.28	0.36
3	0.76	0.82	0.86	0.79	0.75	0.69
4	0.87	1.03	0.85	0.90	0.89	0.96
5 (High)	0.99	1.06	1.06	1.22	1.05	1.42
High-Low	0.64 (2.14)	0.72 (2.44)	0.77 (2.77)	1.01 (3.74)	0.81 (3.79)	1.29 (4.43)
CAPM Alpha	0.64 (2.13)	0.70 (2.34)	0.77 (2.77)	1.02 (3.75)	0.81 (3.79)	1.31 (4.45)
Annual SR	0.73	0.82	0.92	1.24	1.47	1.64

Table A.6 Predicting 30-minute ahead ETF returns: *ElasticNet* Predictors

This table reports the estimated regression coefficients and Newey-West t -statistics (in parentheses) from Fama-MachBeth cross-sectional regressions predicting every intraday half-hour ETF return in interval i ($i = 2, \dots, 13$) on day d by the *ElasticNet* and *News* signals calculated using inputs from interval $i - 1$, where *ElasticNet* is the trading signal constructed by using the ElasticNet method in Eq. (24). Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. *News* is the news-driven signal from Eq. (10). *RET* is the lagged overnight or 30-minute ETF return. *ILLQ* is the natural logarithm of the average daily ratio of the absolute ETF return to the dollar trading volume over the five-day period preceding each day. *RVOL* is the annualized realized volatility, defined as the square root of the annualized realized variance, which is 252 times the sum of squared five-minute intraday ETF returns within each trading day. *Mom* for a given day d is the cumulative ETF return from day $d-252$ to day $d-21$. *Size* is the natural logarithm of the net asset value of the ETF, estimated by the product of the closing price and the number of shares outstanding from CRSP and is updated daily. The last column reports the average coefficient, t -statistic, and adjusted R^2 across all return intervals.

Return Interval													
Start	10:00	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	Mean
End	10:30	11:00	11:30	12:00	12:30	13:00	13:30	14:00	14:30	15:00	15:30	16:00	
Intercept	-0.317 (-0.29)	1.210 (1.23)	1.491 (1.76)	0.329 (0.45)	0.642 (0.89)	0.649 (0.91)	-0.485 (-0.68)	-0.314 (-0.46)	-1.053 (-1.32)	-1.745 (-1.12)	1.485 (1.92)	-1.864 (-2.01)	0.002 (0.11)
<i>ElasticNet</i>	0.099 (6.27)	0.192 (10.36)	0.199 (12.93)	0.160 (11.09)	0.210 (13.60)	0.223 (14.56)	0.199 (12.03)	0.184 (10.97)	0.156 (9.34)	0.148 (11.13)	0.145 (7.74)	0.117 (6.98)	0.169 (10.59)
<i>News</i>	0.019 (21.54)	0.107 (1.58)	-0.057 (-0.71)	0.069 (1.59)	0.019 (0.81)	0.013 (1.28)	0.048 (1.19)	-0.069 (-1.36)	0.107 (1.02)	0.217 (1.08)	0.009 (0.69)	0.534 (0.86)	0.085 (2.46)
RET	-0.078 (-26.16)	-0.087 (-20.98)	-0.103 (-25.03)	-0.104 (-23.81)	-0.119 (-28.13)	-0.130 (-32.73)	-0.138 (-30.54)	-0.147 (-31.53)	-0.161 (-31.51)	-0.157 (-35.68)	-0.164 (-35.85)	-0.208 (-40.19)	-0.133 (-30.18)
ILLIQ	-0.013 (-0.15)	0.042 (0.66)	0.020 (0.35)	-0.017 (-0.36)	0.033 (0.68)	0.093 (1.96)	-0.028 (-0.55)	-0.010 (-0.22)	-0.037 (-0.74)	-0.134 (-1.23)	0.028 (0.48)	-0.081 (-1.31)	-0.009 (-0.04)
RVOL	-1.380 (-0.92)	-0.006 (-0.00)	-2.276 (-1.91)	-1.707 (-1.69)	-0.367 (-0.37)	0.107 (0.12)	-1.676 (-1.77)	-1.865 (-2.11)	-0.450 (-0.44)	0.872 (0.93)	0.110 (0.12)	1.232 (1.24)	-0.617 (-0.57)
Mom	0.439 (0.98)	0.126 (0.34)	-0.118 (-0.38)	0.086 (0.29)	-0.570 (-2.11)	0.498 (1.79)	-0.510 (-2.04)	-0.028 (-0.12)	0.193 (0.74)	0.374 (1.53)	-0.181 (-0.72)	0.159 (0.52)	0.039 (0.07)
Size	-0.002 (-0.01)	-0.119 (-1.06)	-0.097 (-0.96)	-0.084 (-0.90)	0.002 (0.03)	0.182 (2.25)	0.005 (0.06)	0.002 (0.02)	0.000 (-0.00)	-0.135 (-0.96)	-0.101 (-1.02)	-0.042 (-0.36)	-0.032 (-0.24)
Adj- R^2 (%)	15.56	15.04	15.10	14.84	14.21	13.79	14.37	14.70	15.21	15.75	15.72	14.69	14.92

Table A.7 Performance of long-short portfolios sorted on the basis of different overnight LASSO signals: day-of-week effect

This table reports the performance of the long-short portfolios formed on the basis of different overnight LASSO signals on each day of the week. At 10:00 of each trading day d , we sort stocks into quintile portfolios based on their overnight $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys stocks in the top quintile with high signal values and sells stocks in the bottom quintile with low signal values, with a holding period from 10:00 on day d until market close of the same day. Panel A reports the average long-short portfolio returns on each day of the week, and Panel B reports the corresponding annual Sharpe ratios. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All overnight LASSO signals correspond to interval $i = 1$ in Eq. (9). $LASSO$ is the raw LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). We multiply daily returns by 21 to get monthly returns in percentage.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: Average Return</i>						
Mon	0.62 (2.84)	0.82 (3.90)	0.81 (3.56)	0.83 (3.91)	0.80 (3.90)	1.21 (4.35)
Tue	0.62 (2.20)	0.72 (3.44)	0.67 (2.43)	0.65 (2.76)	0.68 (3.94)	1.03 (4.72)
Wed	0.72 (2.78)	0.78 (3.21)	0.79 (3.39)	0.79 (3.32)	0.79 (4.13)	1.31 (5.04)
Thu	0.59 (2.32)	0.62 (3.55)	0.71 (2.73)	0.65 (2.78)	0.69 (4.12)	1.27 (4.40)
Fri	0.63 (2.22)	0.68 (3.32)	0.73 (3.01)	0.74 (3.32)	0.73 (3.98)	1.13 (4.23)
<i>Panel B: Annual Sharpe Ratio</i>						
Mon	0.98	1.36	1.75	1.37	1.62	2.36
Tue	1.20	1.41	1.36	1.47	1.95	2.34
Wed	1.36	1.53	1.81	1.73	2.10	2.50
Thu	1.40	1.51	1.31	1.56	1.19	2.56
Fri	1.17	1.51	1.57	1.41	1.57	2.06

Table A.8 Performance of quintile portfolios sorted on the basis of different overnight LASSO signals: 2008 financial crisis and 2020 COVID-19

This table reports the performance of quintile portfolios formed on the basis of different overnight LASSO signals during the 2008 financial crisis between December 2007 and July 2009 in Panel A, and during 2020 COVID-19 period between January 2020 and July 2020 in Panel B. At 10:00 of each trading day d , we sort ETFs into quintile portfolios based on their overnight $LASSO$, $LASSO_{ADP}$, $LASSO_{CS}$, and $LASSO_{LF}$ signals along with two composite signals. Then we compute equal-weighted returns on each quintile portfolio and a spread portfolio that buys ETFs in the top quintile with high signal values and sells ETFs in the bottom quintile with low signal values, with a holding period from 10:00 on day d until market close of the same day. Our ETF sample consists of all sector ETFs with monthly holding data from the CRSP Mutual Fund database. All overnight LASSO signals correspond to interval $i = 1$ in Eq. (9). $LASSO$ is the simple LASSO signal, $LASSO_{ADP}$ is the adaptive LASSO signal, $LASSO_{CS}$ is the LASSO signal based on customer-supplier relation, and $LASSO_{LF}$ is the LASSO signal based on leading firms. The first composite signal $Composite^{AVG}$ is constructed by simply averaging the four LASSO signals. The second composite signal $Composite^{GBRT}$ combines the four signals using Gradient Boosted Regression Trees (GBRT). We multiply daily returns by 21 to get monthly returns in percentage. The row labeled “High-Low” reports the average monthly return spread between Quintiles 5 and 1, and the row labeled “CAPM Alpha” reports the monthly return spread adjusted by market exposure. The row labeled “Annual SR” reports the annualized Sharpe ratio for each long-short portfolio.

	$LASSO$	$LASSO_{ADP}$	$LASSO_{CS}$	$LASSO_{LF}$	$Composite^{AVG}$	$Composite^{GBRT}$
<i>Panel A: 2008 Financial Crisis</i>						
1 (Low)	-0.38	-0.14	-0.14	-0.85	-0.30	-1.15
2	0.11	0.48	-0.47	-0.17	0.01	-0.11
3	1.07	0.24	-0.27	0.78	0.87	1.32
4	1.17	0.93	1.10	0.60	0.17	0.29
5 (High)	0.16	0.53	1.78	1.71	1.43	1.79
High-Low	0.54	0.66	1.92	2.56	1.73	2.94
	(0.54)	(0.68)	(2.03)	(2.63)	(1.74)	(2.97)
CAPM Alpha	0.55	0.66	1.93	2.57	1.74	2.96
	(0.54)	(0.68)	(2.06)	(2.66)	(1.75)	(3.05)
Annual SR	0.54	0.71	2.11	2.85	1.84	2.94
<i>Panel B: 2020 COVID-19</i>						
1 (Low)	-0.51	-0.77	-1.66	-0.91	-1.32	-1.49
2	0.18	0.60	0.70	1.03	0.59	0.80
3	-0.15	0.31	0.68	0.40	0.66	0.49
4	1.27	0.61	1.16	0.70	0.68	0.65
5 (High)	1.75	1.81	1.67	1.33	1.93	2.10
High-Low	2.26	2.58	3.33	2.24	3.25	3.59
	(3.36)	(3.31)	(2.66)	(2.85)	(4.34)	(4.19)
CAPM Alpha	2.27	2.61	3.30	2.20	3.21	3.59
	(3.37)	(3.35)	(2.70)	(2.84)	(4.44)	(4.27)
Annual SR	3.87	4.75	4.13	4.31	4.95	5.51